



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

FULL SYLLABUS

16-02-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-I)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	C	B	D		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,D	B,C,D	A,B,C	A,D	A,C	A,C
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	0.50	7.50	1.33	0.67	1.73	3.00
SECTION-II (ii)	Q.	7	8	9			
	A.	4	49	9			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	B	B	A		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	C,D	A,D	B,C	A,B,D	A,C,D	A,B
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	1.62	3.20 or 3.90-3.91	166.28	141.33 or 141.34	100.10	2.50
SECTION-II (ii)	Q.	7	8	9			
	A.	9,13	8,12	6			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	A	A	C		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B,D	A,B	A,B,C	A,B,C	A,B	B,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	76.25	24.50	1.00	1.50	9.00	77.14
SECTION-II (ii)	Q.	7	8	9			
	A.	4	2	7			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. **Ans (C)**

Given 10 VSD = 9 MSD

$$1 \text{ VSD} = \frac{9}{10} \text{ MSD}$$

Here MSD → Main Scale division

VSD → Vernier Scale division

Least count = 1 MSD – 1 VSD

$$= \left(1 - \frac{9}{10}\right) \text{ MSD}$$

$$= 0.1 \text{ MSD}$$

$$= 0.1 \times 0.1 \text{ cm}$$

$$= 0.01 \text{ cm}$$

As '0' of V.S. lie before '0' of M.S.

$$\text{Zero error} = -[10 - 6] \text{ L.C.}$$

$$= -4 \times 0.01 \text{ cm}$$

$$= -0.04 \text{ cm}$$

$$\text{Reading} = 3.1 \text{ cm} + 1 \times \text{LC}$$

$$= 3.4 \text{ cm} + 1 \times 0.01 \text{ cm}$$

$$= 3.11 \text{ cm}$$

$$\text{True diameter} = \text{Reading} - \text{Zero error}$$

$$= 3.11 - (-0.04) \text{ cm} = 3.15 \text{ cm}$$

2. Ans (C)

Process 1

P = constant, Volume increases and temperature also increases

⇒ W = positive, ΔU = positive

⇒ Heat is positive and supplied to gas

Process -2

V = constant, Pressure decrease

⇒ Temperature decreases

$$W = \int p dV = 0$$

$$\Delta T \text{ is negative and } \Delta U = \frac{f}{2} nR\Delta T$$

⇒ ΔU is negative

$$\Delta Q = \Delta U + W$$

ΔQ → Heat is negative and rejected by gas

Process 3

P = constant, Volume decreases

⇒ Temperature also decreases

$$W = P\Delta V = \text{negative}$$

$$\Delta U = \frac{f}{2} nR\Delta T = \text{negative}$$

$$\Delta Q = W + \Delta U = \text{negative}$$

Heat is negative and rejected by gas.

Process 4

V = constant, Pressure increases

$$W = \int p dV = 0$$

PV = nRT ⇒ Temperature increase

$$\Rightarrow \Delta U = \frac{f}{2} nR\Delta T \text{ is positive}$$

$$\Delta Q = \Delta U + W$$

= positive

Ans. (C) step 1 and step 4

3. Ans (B)

Focal length of convex lens (f_1)

$$\frac{1}{f_1} = (\mu - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

$$= (1.5 - 1) \left[\frac{1}{20} - \left(\frac{1}{-20} \right) \right]$$

$$\frac{1}{f_1} = \frac{1}{20}$$

$$\Rightarrow f_1 = +20 \text{ cm}$$

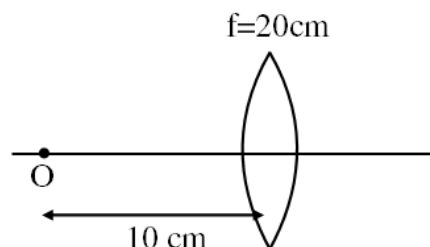
Focal length of concave lens (f_2)

$$\frac{1}{f_2} = (\mu - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

$$\frac{1}{f_2} = (1.5 - 1) \left[-\frac{1}{20} - \frac{1}{20} \right] = \frac{1}{-20}$$

$$\Rightarrow f_2 = -20 \text{ cm}$$

For lens 1

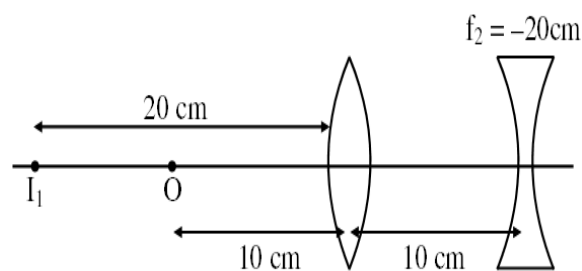


$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow v = -20 \text{ cm}$$

$$m_1 = \frac{v}{u} = \frac{-20}{-10} = 2$$

For lens 2



$$u = -30, f = -20, \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$v = -12 \text{ cm}$$

$$m_2 = \frac{v}{u} = \frac{-12}{-30} = \frac{2}{5}$$

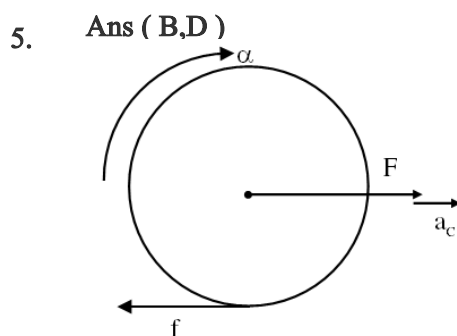
Net magnification

$$m = m_1 m_2 = 2 \times \frac{2}{5} = \frac{4}{5} = 0.8$$

4. **Ans (D)**
Out of 1000 nuclei of Q 60% may go α -decay
 $\Rightarrow$ 600 nuclei may have α -decay
 $\lambda = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{20}$
 $t = 1 \text{ hour} = 60 \text{ minutes}$
Using
 $N = N_0 e^{-\lambda t} = 600 \times e^{-\frac{\ln 2}{20} \times 60}$
 $N = 75$
 $\Rightarrow$ 75 Nuclei are left after one hour
So, No. of nuclei decayed
 $= 600 - 75 = 525$

PART-1 : PHYSICS

SECTION-I (ii)



$$F - f = ma_c$$

$$fR = I_C \alpha$$

$$a_c - \alpha R = 0$$

$$F - I_C \frac{\alpha}{R} = ma_c$$

$$a_c = \frac{F}{\frac{I_C}{R^2} + m}$$

$$f = \frac{I_C \alpha}{R} = \frac{I_C}{R^2} a_c = \frac{I_C}{R^2} \frac{F}{\left[\frac{I_C}{R^2} + m \right]}$$

$$f = \frac{F}{\left(m + \frac{I_C}{R^2} \right)}$$

Thin walled hollow cylinder

$$I_C = mR^2$$

$$a_c = \frac{F}{2m}$$

$$fR = I_C \alpha = \frac{I_C a_c}{R}$$

$$f = \frac{I_C a_c}{R^2} \leq \mu mg$$

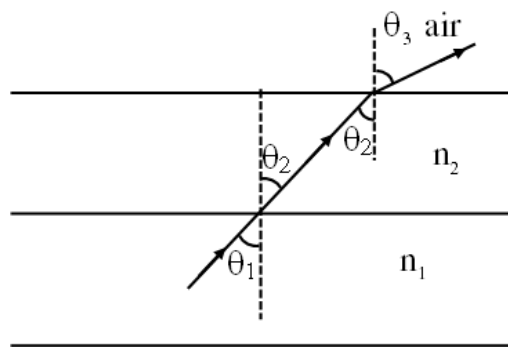
$$a_c \leq \frac{\mu mg R^2}{I_C}$$

$$\text{for solid cylinder } I_C = \frac{mR^2}{2}$$

$$a_c \leq 2\mu g$$

$$(a_c)_{\max} = 2\mu g$$

6. **Ans (B,C,D)**



$$\sin \theta > \frac{1}{n_1} \text{ (Given)}$$

$$\text{i.e. } \sin \theta_1 > \frac{1}{n_1}$$

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$\sin \theta_2 = \frac{n_1 \sin \theta_1}{n_2}$$

$$\text{If } n_1 = n_2 \text{ then } \theta_2 = \theta_1$$

$$n_2 \sin \theta_2 = (1) \sin \theta_3$$

$$\sin \theta_3 = n_2 \sin \theta_2$$

$$\sin \theta_3 = n_1 \sin \theta_1$$

$$\sin \theta_1 = \frac{\sin \theta_3}{n_1} > \frac{1}{n_1}$$

$$\sin \theta_3 > 1$$

$$\theta_3 > 90^\circ$$

This means ray cannot enter air

$$\text{For } n_1 > n_2; \sin \theta_1 = \frac{n_2}{n_1} \sin \theta_2 > \frac{1}{n_1}$$

$$\sin \theta_2 > \frac{1}{n_2}$$

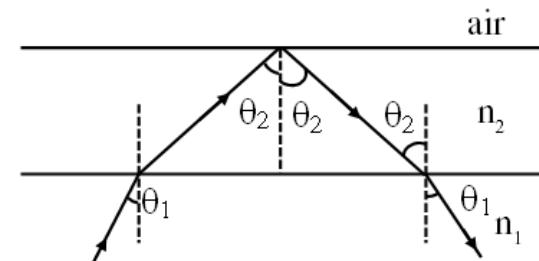
for surface 2 – air interface

$$n_2 \sin \theta_2 = \sin \theta_3$$

$$\sin \theta_2 = \frac{\sin \theta_3}{n_2} > \frac{1}{n_2}$$

$$\theta_2 > 90^\circ$$

It means ray is reflected back in medium-2



for surface 1 – surface 2 interface

$$n_2 \sin \theta_2 = n_1 \sin \theta_1$$

$$\sin \theta_{2C} = \frac{n_1}{n_2}$$

$$\theta_{2C} : \text{critical angle}$$

for ray to enter medium-1

$$\theta_2 < \theta_{2c}$$

$$\sin \theta_2 < \sin \theta_{2c}$$

$$\frac{n_1}{n_2} \sin \theta_1 < \frac{n_1}{n_2}$$

$$\sin \theta_1 < 1$$

$$\theta_1 < 90^\circ, \text{ which is true}$$

Hence ray enters medium-1

$$\text{For } n_2 > n_1$$

$$\frac{n_2}{n_1} \sin \theta_2 > \frac{n_2}{n_1}$$

$$\sin \theta_2 > \frac{1}{n_2}$$

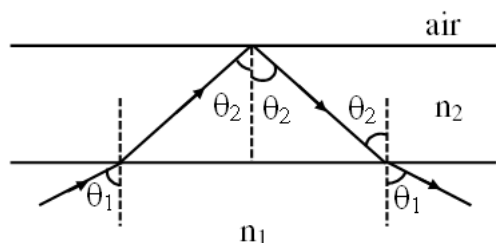
For surface 2 – air interface

$$n_2 \sin \theta_2 = \sin \theta_3$$

$$\sin \theta_2 = \frac{\sin \theta_3}{n_2} > \frac{1}{n_2}$$

$$\theta_2 > 90^\circ$$

It means ray is reflected back in medium - 2



$$n_2 \sin \theta_2 = n_1 \sin \theta_1$$

$$\sin \theta_1 = \frac{n_2}{n_1} \sin \theta_2$$

$$\sin \theta_{2c} = \frac{n_1}{n_2}; \theta_{2c} \rightarrow \text{critical angle}$$

For ray to enter medium - 1

$$\theta_2 < \theta_{2c}$$

$$\sin \theta_2 < \sin \theta_{2c}$$

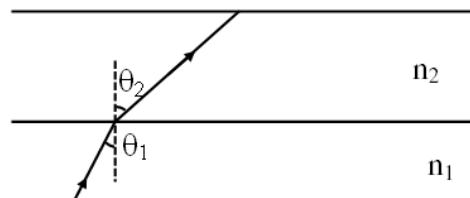
$$\frac{n_1}{n_2} \sin \theta_1 < \frac{n_1}{n_2}$$

$$\sin \theta_1 < 1$$

$$\theta_1 < 90^\circ, \text{ which is true}$$

Hence ray enters medium - 1

$$\text{Let } n_2 = 1$$



$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$n_2 = 1$$

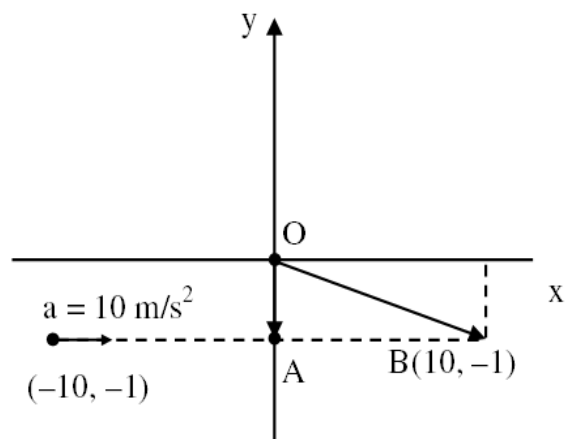
$$n_1 \sin \theta_1 = \sin \theta_2$$

$$\sin \theta_1 = \frac{\sin \theta_2}{n_1} > \frac{1}{n_1}$$

$$\sin \theta_2 > 1 \Rightarrow \theta_2 > 90^\circ$$

ray is reflected back in medium -

7. Ans (A,B,C)



$$\vec{r}_A = -\hat{j}$$

$$S = \frac{1}{2}at^2$$

$$20 = \frac{1}{2} \times 10 \times t^2$$

$$t = 2 \text{ sec}$$

$$\vec{\tau}_0 = \vec{r} \times \vec{F}; \vec{r}_B = 10\hat{i} - \hat{j}$$

$$\vec{F} = m\vec{a} = 0.2 \times 10\hat{i} = 2\hat{i}$$

$$\vec{\tau}_0 = (10\hat{i} - \hat{j}) \times (2\hat{i})$$

$$\vec{\tau}_0 = 2\hat{k}$$

$$\vec{L}_0 = \vec{r}_B \times \vec{p} = \vec{r}_B \times m\vec{v}$$

$$\vec{v} = \vec{a}t = 10\hat{i} \times 2 = 20\hat{i}$$

$$\vec{L}_0 = (0.2) [(10\hat{i} - \hat{j}) \times 20\hat{i}] = 4\hat{k}$$

At point A(0, -1)

$$\vec{\tau}_0 = \vec{r}_A \times \vec{F} = (-\hat{j}) \times 2\hat{i} = 2\hat{k}$$

8. Ans (A,D)

For A

When the transition is from any level to $n = 2$, then photon emitted belong to Balmer series.

∴ For longest wavelength, transition occurs from $n = 3$ to $n = 2$.

$$\therefore \frac{hc}{\lambda_{\max}} = RCh \left[\frac{1}{2^2} - \frac{1}{3^2} \right] \quad \& \quad \text{for shortest}$$

wavelength transition occurs from $n = \infty$ to $n = 2$

$$\therefore \frac{hc}{\lambda_{\min}} = RCh \left[\frac{1}{2^2} - \frac{1}{\infty^2} \right]$$

$$\therefore \frac{\lambda_{\text{longest}}}{\lambda_{\text{shortest}}} = \frac{9}{5}$$

For (B)

$$\lambda_{\text{longest of Balmer}} = \frac{36}{5R}$$

$$\lambda_{\text{shortest of Paschen}} = \frac{9}{R}$$

Hence these wavelength don't overlap.

For (C)

For Lyman series,

$$\frac{1}{\lambda} = R \left[\frac{1}{1} - \frac{1}{m^2} \right]$$

$$\text{Also } \frac{1}{\lambda_0} = R$$

$$\therefore \frac{1}{\lambda} = \frac{1}{\lambda_0} \left[1 - \frac{1}{m^2} \right] \Rightarrow \lambda = \frac{\lambda_0}{1 - \frac{1}{m^2}}$$

For (D)

$$\lambda_{\text{longest of Lyman}} = \frac{4}{3R},$$

$$\lambda_{\text{shortest of Balmer}} = \frac{4}{R}$$

Hence that wavelength don't overlap.

9. Ans (A,C)

EMF developed across the emf of semi-circular rod =

$$\int_1^4 \frac{\mu_0 i}{2\pi r} dr = \frac{\mu_0 i v}{2\pi} \ln 4 = \frac{\mu_0 i v}{\pi} \ln 2$$

From given value,

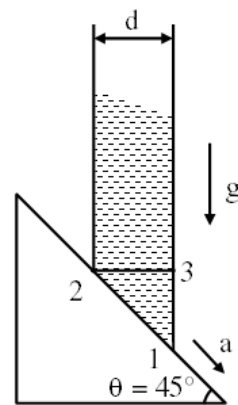
$$E = \frac{4\pi \times 10^{-7} \times 2 \times 3 \times 0.7}{\pi} = 24 \times 7 \times 10^{-8}$$

$$i_{\max} = \frac{E}{R} = \frac{24 \times 7 \times 10^{-8}}{1.4} = 1.2 \times 10^{-6} \text{ A}$$

$$Q_{\max} = C_0 E = 24 \times 7 \times 10^{-8} \times 5 \times 10^{-6} = 8.4 \times 10^{-12} \text{ C}$$

10.

Ans (A,C)



$$\therefore P_1 - P_3 = \rho \left(g - \frac{a}{\sqrt{2}} \right) d$$

$$P_2 - P_3 = \rho \frac{a}{\sqrt{2}} d$$

$$\therefore P_1 - P_2 = \rho d \left[g - \frac{2a}{\sqrt{2}} \right]$$

$$\therefore \frac{P_1 - P_2}{\rho g d} = \left[1 - \sqrt{2} \frac{a}{g} \right] = \beta$$

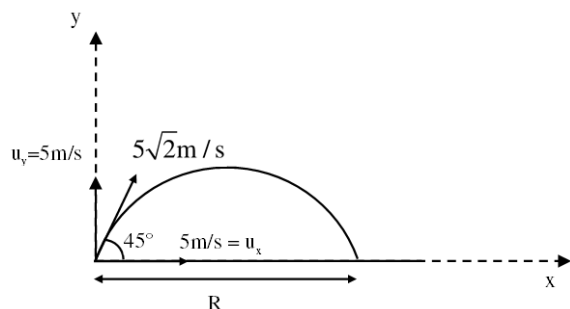
$$\therefore \text{if } \beta = 0, a = \frac{g}{\sqrt{2}} \quad \dots (A)$$

$$\beta = \frac{\sqrt{2} - 1}{2}, a = \frac{g}{2} \quad \dots (C)$$

PART-1 : PHYSICS

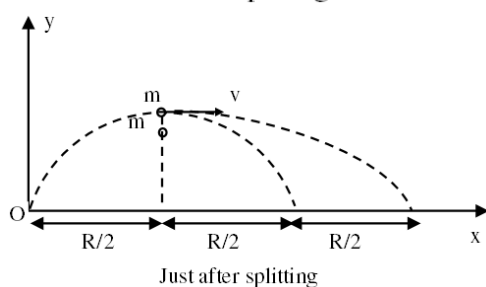
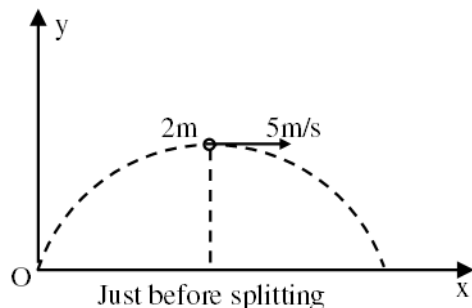
SECTION-II (i)

1. Ans (0.50)



$$\text{Range } R = \frac{2u_x u_y}{g} = \frac{2 \times 5 \times 5}{10} = 5\text{m}$$

$$\text{Time of flight } T = \frac{2u_y}{g} = \frac{2 \times 5}{10} = 1 \text{ sec}$$



$\therefore$ Time of motion of one part falling vertically downwards is $= 0.5 \text{ sec} = \frac{T}{2}$

$\Rightarrow$ Time of motion of another part, $t = \frac{T}{2} = 0.5 \text{ sec}$

From momentum conservation $\Rightarrow P_i = P_f$

$$2m \times 5 = m \times v$$

$$v = 10 \text{ m/s}$$

Displacement of other part in 0.5 sec in horizontal

$$\text{direction} = v \frac{T}{2}$$

$$= 10 \times 0.5 = 5 \text{ m} = R$$

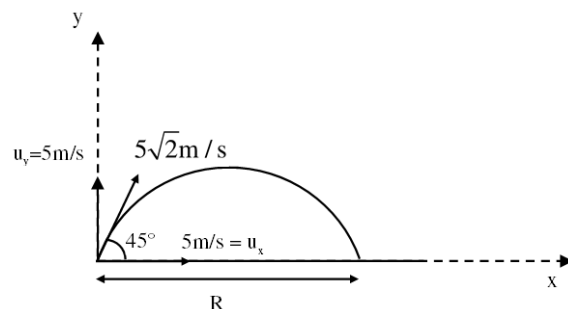
$\therefore$ Total distance of second part from point 'O' is,

$$x = \frac{3R}{2} = 3 \times \frac{5}{2}$$

$$x = 7.5 \text{ m}$$

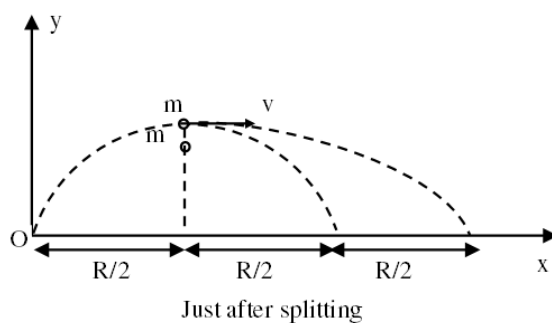
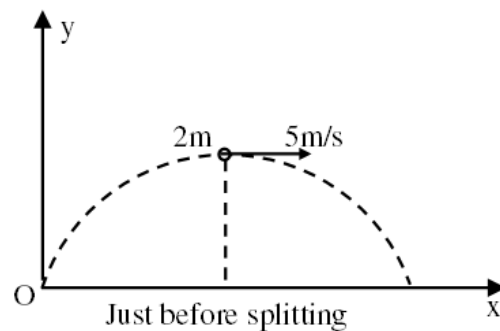
$$\Rightarrow t = 0.5 \text{ sec}$$

2. Ans (7.50)



$$\text{Range } R = \frac{2u_x u_y}{g} = \frac{2 \times 5 \times 5}{10} = 5\text{m}$$

$$\text{Time of flight } T = \frac{2u_y}{g} = \frac{2 \times 5}{10} = 1 \text{ sec}$$



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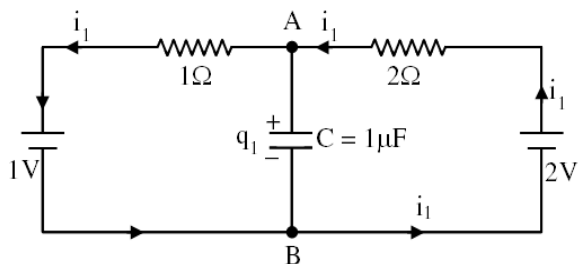
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$$x = 7.5 \text{ m}$$

$$\Rightarrow t = 0.5 \text{ sec}$$

3. Ans (1.33)



Switch connected to position 'P'

$$V_A - 1 \cdot i_1 - 1 + 2 - 2i_1 = V_A$$

$$3i_1 = 1$$

$$i_1 = \frac{1}{3} \text{ A}$$

$$V_A - 1 \cdot i_1 - 1 = V_B$$

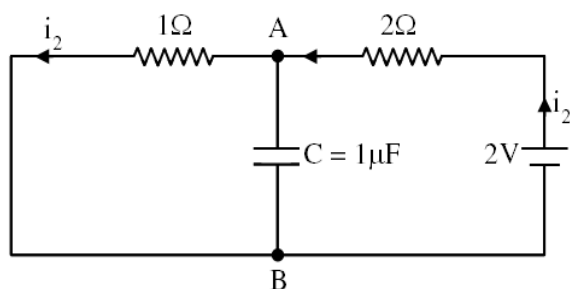
$$V_A - V_B = 1 + i_1 = \frac{4}{3} \text{ volt}$$

$$\text{Potential drop across capacitor } \Delta V = \frac{4}{3} \text{ volt}$$

$$\therefore \text{Charge on capacitor } q_1 = C\Delta V$$

$$= 1 \times \frac{4}{3} \mu\text{C}$$

$$q_1 = 1.33 \mu\text{C}$$



Switch at Position 'Q'

$$V_A - 1 \cdot i_2 + 2 - 2i_2 = V_A$$

$$3i_2 = 2$$

$$i_2 = \frac{2}{3} \text{ A}$$

$$V_A - i_2 \times 1 = V_B$$

$$V_A - V_B = i_2 \times 1 = \frac{2}{3} \text{ volt}$$

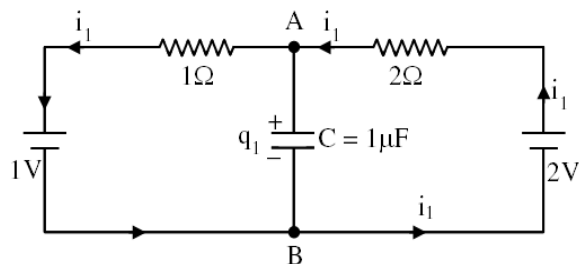
Potential difference across capacitor

$$\Delta V = \frac{2}{3} \text{ volt}$$

$$\therefore \text{Charge on capacitor } q_2 = C\Delta V$$

$$= 1 \times \frac{2}{3} = 0.67 \mu\text{C}$$

4. Ans (0.67)



Switch connected to position 'P'

$$V_A - 1 \cdot i_1 - 1 + 2 - 2i_1 = V_A$$

$$3i_1 = 1$$

$$i_1 = \frac{1}{3} \text{ A}$$

$$V_A - 1 \cdot i_1 - 1 = V_B$$

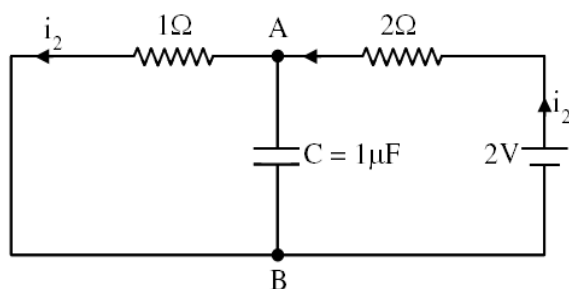
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$$\therefore \text{Charge on capacitor } q_1 = C\Delta V$$

$$= 1 \times \frac{4}{3} \mu\text{C}$$

$$q_1 = 1.33 \mu\text{C}$$



Switch at Position 'Q'

$$V_A - 1 \cdot i_2 + 2 - 2i_2 = V_A$$

$$3i_2 = 2$$

$$i_2 = \frac{2}{3} \text{ A}$$

$$V_A - i_2 \times 1 = V_B$$

$$V_A - V_B = i_2 \times 1 = \frac{2}{3} \text{ volt}$$

Potential difference across capacitor

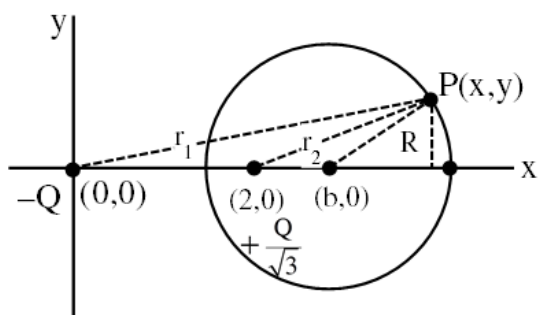
$$\Delta V = \frac{2}{3} \text{ volt}$$

$$\therefore \text{Charge on capacitor } q_2 = C\Delta V$$

$$= 1 \times \frac{2}{3} = 0.67 \mu\text{C}$$

5. Ans (1.73)

Let a point P on circle



$$V_p = 0 = \frac{k(-Q)}{r_1} + \frac{kQ/\sqrt{3}}{r_2}$$

$$\frac{kQ}{r_1} = \frac{kQ/\sqrt{3}}{r_2}$$

$$\frac{1}{\sqrt{x^2 + y^2}} = \frac{1}{\sqrt{3}\sqrt{(x-2)^2 + y^2}}$$

$$3(x-2)^2 + 3y^2 = x^2 + y^2$$

$$3(x^2 + 4 - 4x) - x^2 + 2y^2 = 0$$

$$2x^2 + 12 - 12x + 2y^2 = 0$$

$$x^2 + 6 - 6x + y^2 = 0$$

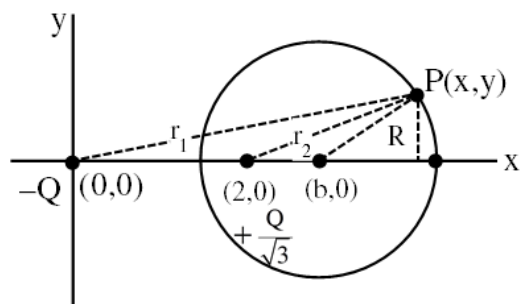
$$(x-3)^2 + y^2 = (\sqrt{3})^2$$

$$R = \sqrt{3} = 1.73,$$

$$b = 3$$

6. Ans (3.00)

Let a point P on circle



$$V_p = 0 = \frac{k(-Q)}{r_1} + \frac{kQ/\sqrt{3}}{r_2}$$

$$\frac{kQ}{r_1} = \frac{kQ/\sqrt{3}}{r_2}$$

$$\frac{1}{\sqrt{x^2 + y^2}} = \frac{1}{\sqrt{3}\sqrt{(x-2)^2 + y^2}}$$

$$3(x-2)^2 + 3y^2 = x^2 + y^2$$

$$3(x^2 + 4 - 4x) - x^2 + 2y^2 = 0$$

$$2x^2 + 12 - 12x + 2y^2 = 0$$

$$x^2 + 6 - 6x + y^2 = 0$$

$$(x-3)^2 + y^2 = (\sqrt{3})^2$$

$$R = \sqrt{3} = 1.73,$$

$$b = 3$$

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (4)

$$r = \frac{mv}{qB} = \frac{\sqrt{2mqV}}{qB}$$

$$\frac{P^2}{2m} = K.E = qV$$

$$\frac{r_s}{r_a} = \sqrt{\frac{32}{1}} \times \frac{2}{4} = 4$$

$$\frac{r_s}{r_a} = 4$$

8. Ans (49)

$$L = \frac{Ma^2}{3}\Omega + M\left(\frac{3a}{4}\right)^2\Omega + \frac{M\left(\frac{a}{4}\right)^2 4\Omega}{2}$$

$$L = \frac{49}{48}Ma^2\Omega$$

$$n = 49$$

9. Ans (9)

$$\sigma AT^4 = -ms \frac{dT}{dt}$$

$$\int_{200}^{100} \frac{dT}{T^4} = \int_0^{t_1} k dt$$

$$\frac{1}{3T^3} \Big|_{200}^{100} = kt_1$$

$$\frac{1}{3} \left(\frac{1}{100^3} - \frac{1}{200^3} \right) = kt_1$$

$$\frac{1}{3T^3} \Big|_{200}^{50} = kt_2$$

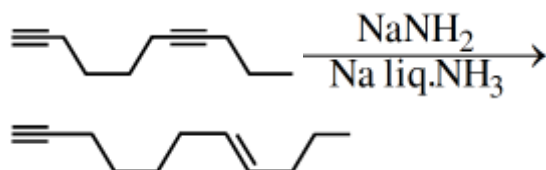
$$\frac{1}{3} \left(\frac{1}{50^3} - \frac{1}{200^3} \right) = kt_2$$

$$\frac{t_2}{t_1} = \left(\frac{200^3 - 50^3}{200^3 - 100^3} \right) \frac{100^3}{50^3} = 9$$

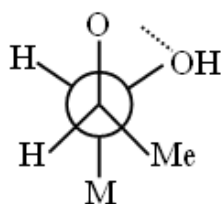
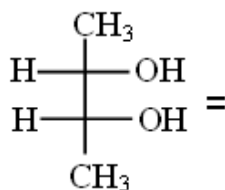
PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (B)



2. Ans (B)



Most stable conformation with H-bonding

Meso butane -2,3,-diol

3. Ans (B)

$$\text{Packing fraction (P.F.)} = \frac{1 \times \frac{4}{3}\pi r_-^3 + 3 \times \frac{4}{3}\pi r_+^3}{a^3}$$

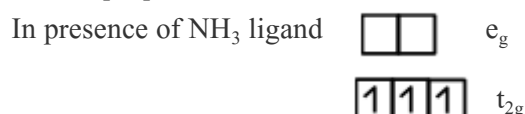
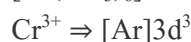
$$\frac{r_+}{r_-} = 0.414 \text{ (square planar void), } a = 2r_-$$

We get,

$$\text{P.F.} = \frac{\frac{4}{3}\pi (r_-^3 + 3r_+^3)}{8r_-^3} = \left[\frac{\pi}{6} \left(1 + 3(0.414)^3 \right) \right]$$

$$= 0.63$$

4. Ans (A)



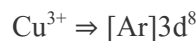
Number of unpaired electrons = 3

$$\mu = \sqrt{n(n+2)} \text{ B.M.}$$

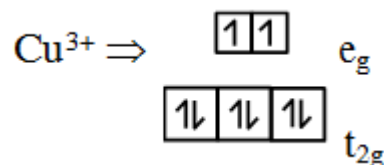
$$\mu = \sqrt{3(3+2)} \text{ B.M.}$$

$$\mu = \sqrt{15} \text{ B.M.}$$

$$\Rightarrow 3.87 \text{ B.M.}$$



In presence of F^- Ligand



Number of unpaired electrons = 2

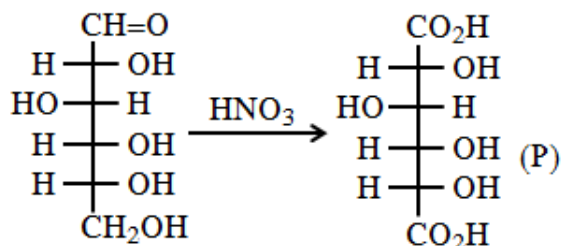
$$\mu = \sqrt{n(n+2)} \text{ B.M.}$$

$$\mu = \sqrt{2(2+2)} \Rightarrow \sqrt{8} \text{ B.M.} \Rightarrow 2.84 \text{ B.M.}$$

PART-2 : CHEMISTRY

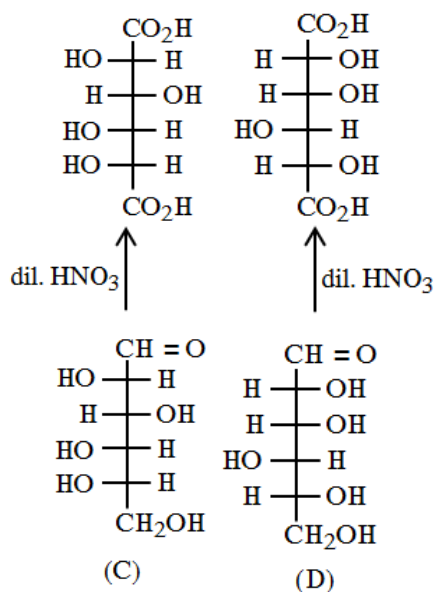
SECTION-I (ii)

5. Ans (C,D)

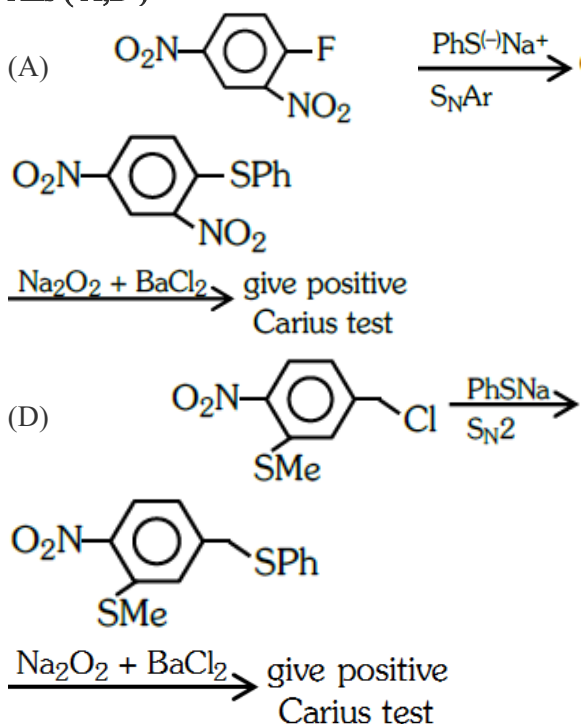


$$[\alpha]_D = 52.7^\circ$$

The enantiomer of P has rotation -52.7° is as follows



6. Ans (A,D)



7. Ans (B,C)

(A) Process of precipitating colloidal solution is called coagulation. Hence false.

(B) For colloidal solutions concentration is very small due to very large molar mass and hence their

colligative properties are very small as compared to true solutions

$\therefore \Delta T_f$ is lesser for colloidal solution. Hence true.

(C) At CMC surfactant form micelles. Hence true

(D) Micelles and macromolecular colloids are two different types of colloids. Hence false.

8. Ans (A,B,D)

From state I to II (Reversible isothermal expansion)

$\Rightarrow P$ decreases, V increases, T constant

H constant & S increases.

From state II to III (Reversible adiabatic expansion)

$\Rightarrow P$ decreases, V increases, T decreases

H decreases, S constant

$\therefore$ Plots (A), (B), (D) are correct while (C) is wrong as from II to III, H is decreasing.

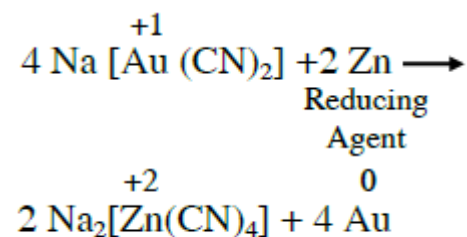
9. Ans (A,C,D)

(A) $\text{PbS} + 2\text{PbO} \rightarrow 3\text{Pb} + \text{SO}_2$ (self reduction)

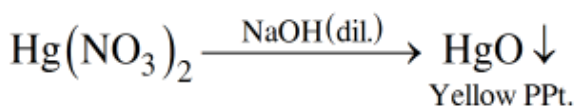
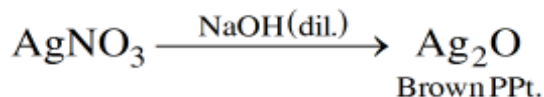
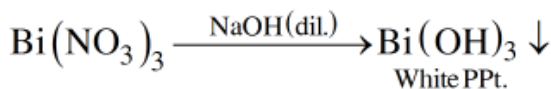
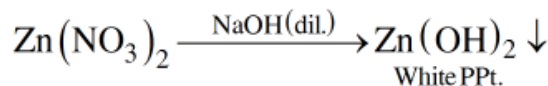
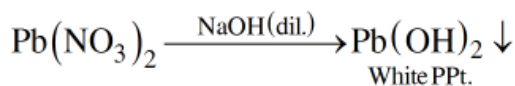
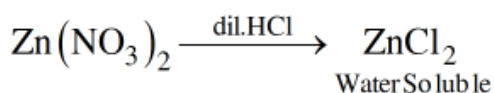
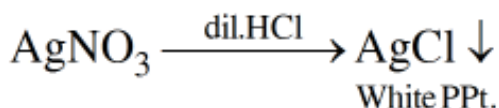
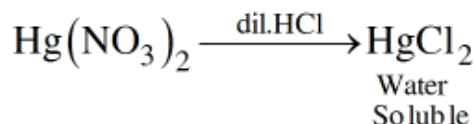
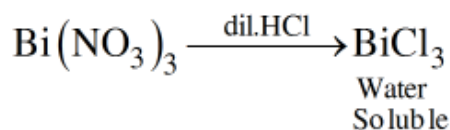
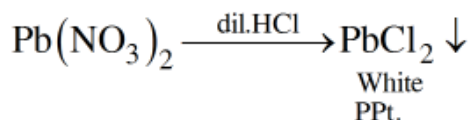
(B) Silica is added to remove impurity of Fe in the form of slag FeSiO_3

(C) CuFeS_2 ore is partially oxidized first by roasting and then self reduction of Cu takes place to produce blister copper.

(D)



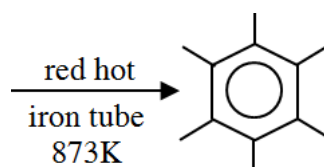
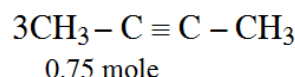
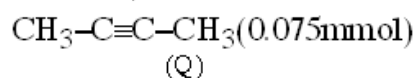
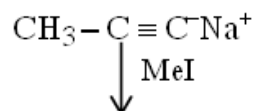
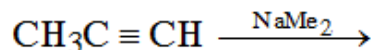
10. Ans (A,B)



PART-2 : CHEMISTRY

SECTION-II (i)

1. Ans (1.62)

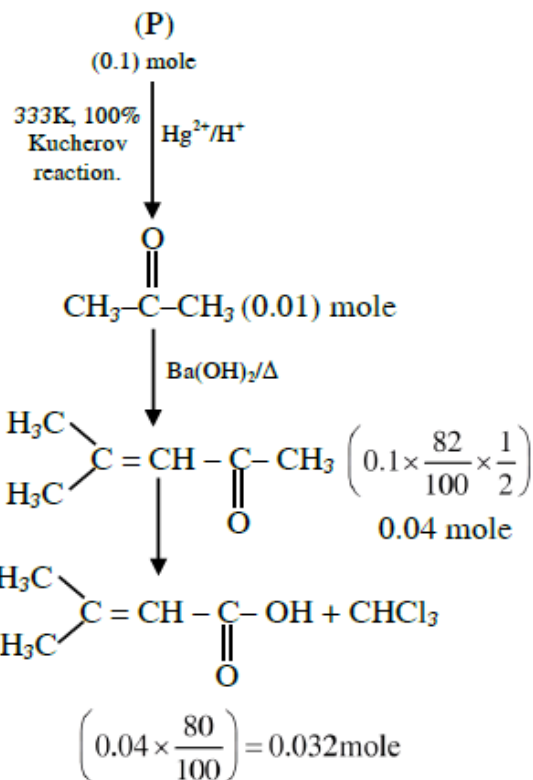


$$(0.075 \times 0.4) \times \frac{1}{3}$$

$$= 0.01 \text{ mole}$$

The value of x = 162 × 0.01 = 1.62 gm

2. Ans (3.20 or 3.90-3.91)



$$60 + 32 + 8 = 100$$

The value of Y = 0.032 × 100 = 3.2

3. Ans (166.28)

$$\Delta G^\circ = -RT \ln \left(\frac{P}{1} \right) = \Delta H^\circ - T \Delta S^\circ$$

$$\ln \left(\frac{P}{1} \right) = -\frac{\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R}$$

$$\text{Slope} = -\frac{\Delta H^\circ}{R} = 10^4 \times \left(-\frac{4}{2} \right)$$

$$\Rightarrow \Delta H^\circ = 2 \times 10^4 \times R = 166.28 \text{ kJ/mole}$$

4. Ans (141.33 or 141.34)

$$\text{From the plot when, } \frac{10^4}{T} = 10 \Rightarrow T = 1000 \text{ K}$$

$$\ln \left(\frac{P_2}{1} \right) = -3$$

Substituting in equation :

$$\ln \left(\frac{P_2}{1} \right) = -\frac{\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R}$$

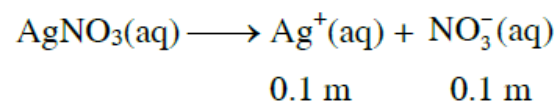
$$\text{We get, } -3 = -\frac{2 \times 10^4 \times R}{R \times 1000} + \frac{\Delta S^\circ}{R}$$

$$\Rightarrow \Delta S^\circ = 17R$$

$$\Rightarrow \Delta S^\circ = 17 \times 8.314 \text{ J/K-mol}$$

$$\Rightarrow \Delta S^\circ = 141.34 \text{ J/K-mol}$$

5. Ans (100.10)



$$\Delta T_b = 0.2 \times 0.5 = 0.1^\circ\text{C} = 0.1 \text{ K}$$

$$\text{Boiling point of solution} = 100.1^\circ\text{C} = X$$

6. Ans (2.50)

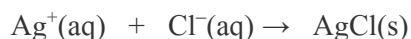


$$0.05 \text{ m} \quad 0.05 \text{ m} \quad 0.05 \text{ m}$$



$$0.05 \text{ m} \quad 0.05 \text{ m} \quad 0.1 \text{ m}$$

Ag^+ and Cl^- combine to form AgCl precipitate



$$t = 0 \quad 0.05 \text{ m} \quad 0.1 \text{ m}$$

$$t = \infty \quad 0 \quad 0.05 \text{ m}$$

In final solution total concentration of all ions :

$$[\text{Cl}^-] + [\text{NO}_3^-] + [\text{Ba}^{2+}] = 0.05 + 0.05 + 0.05$$

$$= 0.15 \text{ m}$$

$$\Delta T_b = 0.5 \times 0.15 = 0.075^\circ\text{C}$$

$$\text{B.P. of solution 'B'} = 100.075^\circ\text{C}$$

$$\text{B.P. of solution 'A'} = 100.1^\circ\text{C}$$

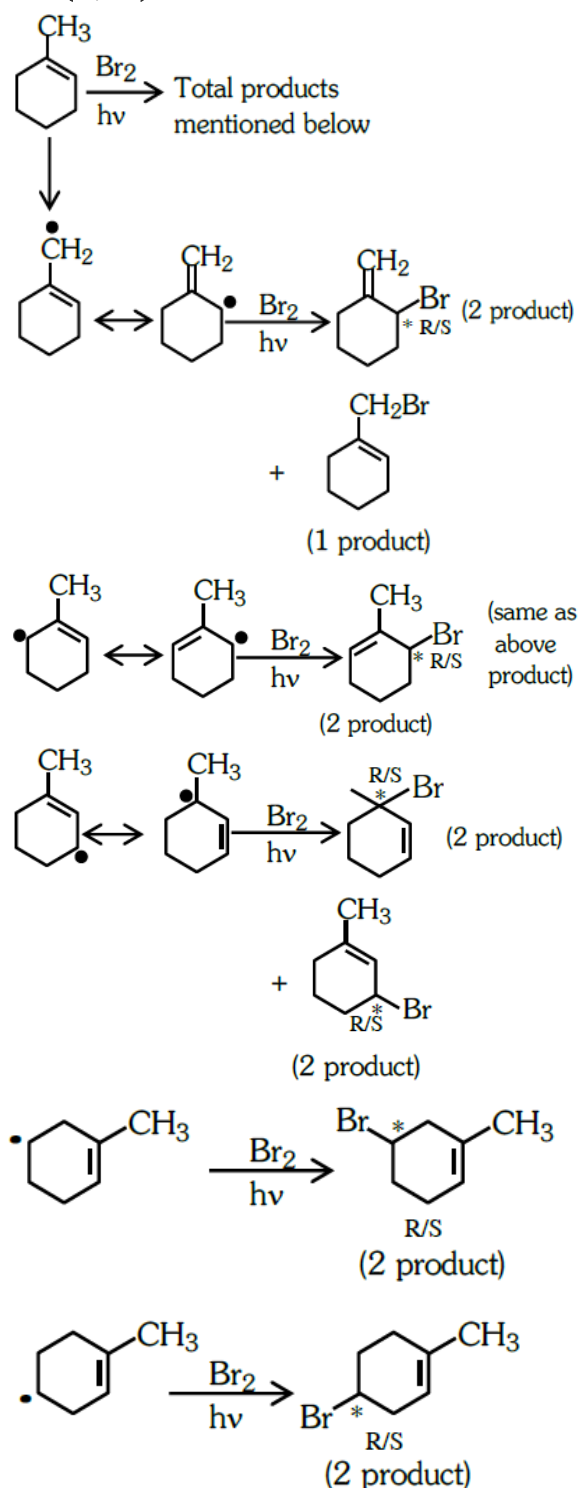
$$|y| = 100.1 - 100.075$$

$$= 0.025 = 2.5 \times 10^{-2}$$

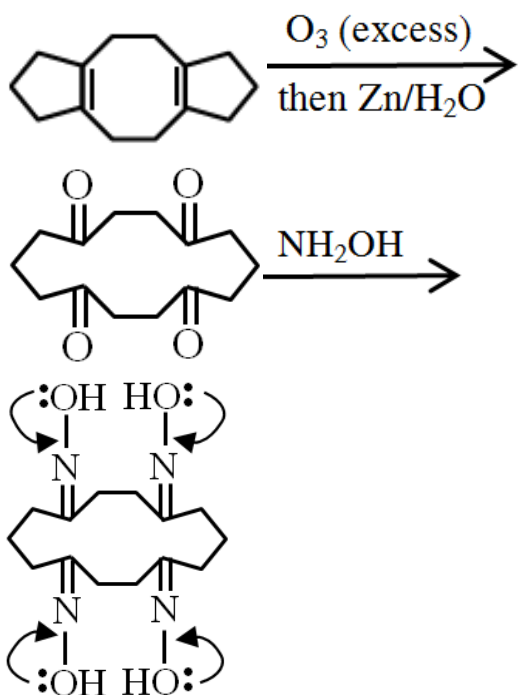
PART-2 : CHEMISTRY

SECTION-II (ii)

7. Ans (9,13)



8. Ans (8,12)



Total 12 atoms are sp^2 hybridised

9. Ans (6)

Isomers

- $$\begin{aligned} \text{(I)} \quad & [\text{Pt}(\text{NH}_3)_4 \text{Cl}_2] \text{Br}_2 \Rightarrow \text{G.I.} = 2 \\ \text{(II)} \quad & [\text{Pt}(\text{NH}_3)_4 \text{Br}_2] \text{Cl}_2 \Rightarrow \text{G.I.} = 2 \\ \text{(III)} \quad & [\text{Pt}(\text{NH}_3)_4 \text{BrCl}] \text{Br} \cdot \text{Cl} \Rightarrow \text{G.I.} = 2 \end{aligned}$$

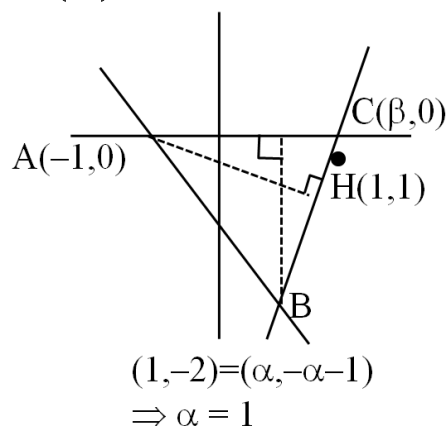
I, II, III are ionisation isomers of each other, each having 2 geometrical isomers.

Total possible isomers will be 6

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B)



one of the vertex is intersection of x-axis and
 $x + y + 1 = 0 \Rightarrow A(-1,0)$

Let vertex B be $(\alpha, -\alpha-1)$

Line $AC \perp BH \Rightarrow \alpha = 1 \Rightarrow B(1, -2)$

Let vertex C be $(\beta, 0)$

Line $AH \perp BC$

$$m_{AH}.m_{BC} = -1$$

$$\frac{1}{2} \cdot \frac{2}{\beta - 1} = -1 \Rightarrow \beta = 0$$

Centroid of ΔABC is $\left(0, -\frac{2}{3}\right)$

Now G(centroid) divides line joining circum centre (O) and ortho centre (H) in the ratio 1: 2

Diagram illustrating the mapping of points (h,k) and $(1,1)$ to points O and H respectively, with a point G marked at 1 . The point G is labeled with the coordinates $\left(0, -\frac{2}{3}\right)$.

$$2h + 1 = 0 \quad 2k + 1 = -z$$

$$h = -\frac{1}{2} \quad k = -\frac{3}{2}$$

$$\Rightarrow \text{circum centre is } \left(-\frac{1}{2}, -\frac{3}{2}\right)$$

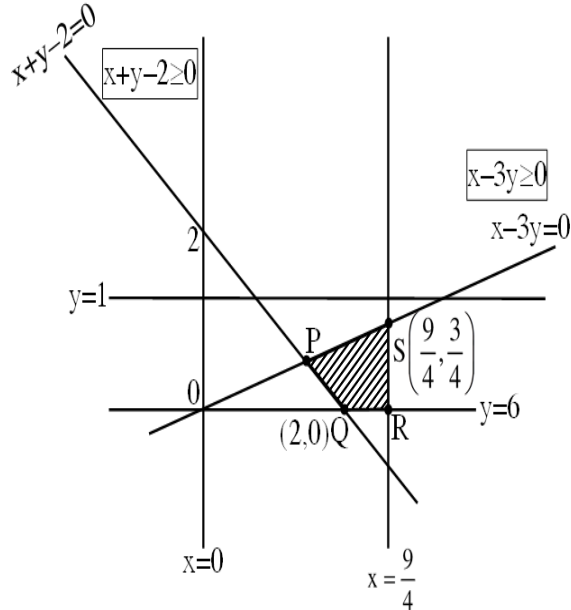
Equation of circum circle is (passing through $C(0,0)$) is

$$x^2 + y^2 + x + 3y = 0$$

2. Ans (A)

$$x + y - 2 = 0 \quad P\left(\frac{3}{2}, \frac{1}{2}\right); \quad Q(2, 0); \quad R\left(\frac{9}{4}, 0\right);$$

$$S\left(\frac{9}{4}, \frac{3}{4}\right)$$



$$\text{Area} = \frac{1}{2} \left| \begin{vmatrix} \frac{3}{2} & \frac{1}{2} \\ 2 & 0 \end{vmatrix} + \begin{vmatrix} 2 & 0 \\ \frac{9}{4} & 0 \end{vmatrix} + \begin{vmatrix} \frac{9}{4} & 0 \\ \frac{9}{4} & \frac{3}{4} \end{vmatrix} + \begin{vmatrix} \frac{9}{4} & \frac{3}{4} \\ \frac{3}{2} & \frac{1}{2} \end{vmatrix} \right|$$

$$= \frac{1}{2} \left| (0-1) + (0-0) + \left(\frac{27}{16} - 0\right) + \left(\frac{9}{8} - \frac{9}{8}\right) \right| = \frac{11}{32}$$

3. Ans (A)

$$P = \frac{P(S_1 \cap (E_1 = E_3))}{P(E_1 = E_3)} = \frac{P(B_{1,2})}{P(B)}$$

$$P(B) = P(B_{1,2}) + P(B_{1,3}) + P(B_{2,3})$$

↓ ↓ ↓
If 1,2 If 1,3 If 2,3
chosen chosen chosen
at start at start at start

$$P(B_{1,2}) = \frac{1}{3} \times \frac{1 \times {}^3C_1}{{}^4C_2} \times \frac{1}{{}^5C_2}$$

1 is definitely chosen from F_2 1,2 chosen from G_2

$$P(B_{1,3}) = \frac{1}{3} \times \frac{1 \times {}^2C_1}{{}^3C_2} \times \frac{1}{{}^5C_2}$$

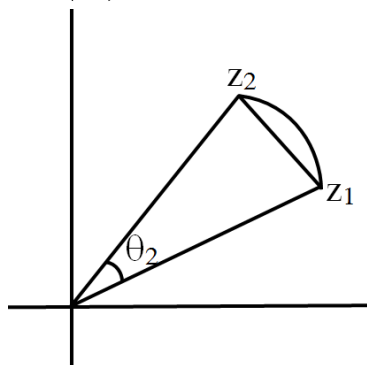
1 is definitely chosen from F_2 1,2 chosen from G_2

$$P(B_{2,3}) = \frac{1}{3} \times \left[\frac{{}^3C_2 \times 1}{{}^4C_2} \times \frac{1}{{}^4C_2} + \frac{1 \times {}^3C_1}{{}^4C_2} \times \frac{1}{{}^5C_2} \right]$$

If 1 is not chosen from F_2 If 1 is chosen from F_2

$$\frac{P(B_{1,2})}{P(B)} = \frac{1}{5}$$

4. Ans (C)



$$|z_1| = |z_2| = \dots |z_{10}| = 1$$

$$\text{angle} = \frac{\text{arc}}{\text{rad}}$$

$$\theta_2 = \text{arc}(z_1 z_2) > (z_2 > z_1)$$

$$P : |z_2 - z_1| + \dots + |z_1 - z_{10}| \leq \theta_1 + \theta_2 + \dots + \theta_{10}$$

$$\Rightarrow |z_2 - z_1| + \dots + |z_1 - z_{10}| \leq 2\pi \text{ P is true}$$

$$z_1^2 = e^{i2\theta_1}, z_k^2 = z_{k-1}^2 \cdot e^{i2\theta_k}$$

$$\text{Let } 2\theta_k = \alpha_k$$

$$z_1^2 = e^{i\alpha_1}, z_k^2 = z_{k-1}^2 \cdot e^{i\alpha_k}$$

$$\alpha_1 + \alpha_2 + \dots + \alpha_k = 4\pi$$

one similar sense

$$|z_1^2 - z_2^2| + \dots + |z_1^2 - z_{10}^2| \leq 4\pi$$

Q is also true

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,B,D)

(A)

$$PEP = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 8 & 13 & 18 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 8 & 13 & 18 \\ 2 & 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 3 & 2 \\ 8 & 18 & 13 \\ 2 & 4 & 3 \end{pmatrix}$$

$$P^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$(B) |EQ + PFQ^{-1}| = |EQ| + |PFQ^{-1}|$$

$$|E| = 0 \text{ and } |F| = 0 \text{ and } |Q| \neq 0$$

$$|EQ| = |E| |Q| = 0, |PFQ^{-1}| = \frac{|P| |F|}{|Q|} = 0$$

$$T = EQ + PFQ^{-1} \quad TQ = EQ^2 + PF = EQ^2 +$$

$$P^2EP = EQ^2 + EP = E(Q^2 + P)$$

$$|TQ| = |E(Q^2 + P)| \Rightarrow |T| |Q| = |E| |Q^2 + P| = 0 \Rightarrow |T| = 0$$

$$(\text{as } |Q| \neq 0) \quad (C) |(EF)^3| > |EF|^2$$

$$\text{Here } 0 > 0 \text{ (false)}$$

$$(D) \text{ as } P^2 = I \Rightarrow P^{-1} = P \text{ so } P^{-1}FP = PFP$$

$$= PPEPP$$

$$= E$$

$$\text{so } E + P^{-1}FP = E + E = 2E$$

$$P^{-1}EP + F \Rightarrow PEP + F = 2PEP$$

$$\text{Tr}(2PEP) = 2\text{Tr}(PEP) = 2\text{Tr}(EPP)$$

$$= 2\text{Tr}(E)$$

6. Ans (A,B)

$$f(x) = \frac{x^2 - 3x - 6}{x^2 + 2x + 4}$$

$$f'(x) = \frac{(x^2 + 2x + 4)(2x - 3) - (x^2 - 3x - 6)(2x + 2)}{(x^2 + 2x + 4)^2}$$

$$f'(x) = \frac{5x(x+4)}{(x^2 + 2x + 4)^2}$$

$$f'(x): \begin{array}{c} + \quad - \quad + \\ -4 \quad 0 \end{array}$$

$$f(-4) = \frac{11}{6}, \quad f(0) = -\frac{3}{2},$$

$$\lim_{x \rightarrow \pm\infty} f(x) = 1$$

$$\text{Range} : \left[-\frac{3}{2}, \frac{11}{6}\right], \text{ clearly } f(x) \text{ is into}$$

7. Ans (A,B,C)

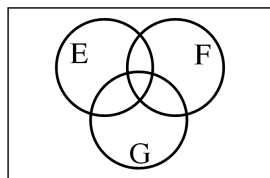
$$P(E) = \frac{1}{8}; P(F) = \frac{1}{6}; P(G) = \frac{1}{4}; P(E \cap F \cap G) = \frac{1}{10}$$

$$(C) P(E \cup F \cup G) = P(E) + P(F) + P(G) - P(E \cap F) - P(F \cap G) - P(G \cap E) + P(E \cap F \cap G)$$

$$= \frac{1}{8} + \frac{1}{6} + \frac{1}{4} - \sum P(E \cap F) + \frac{1}{10}$$

$$= \frac{3+4+6}{24} + \frac{1}{10} - \sum P(E \cap F) = \frac{13}{24} + \frac{1}{10} - \sum P(E \cap F)$$

$$\Rightarrow P(E \cup F \cup G) \leq \frac{13}{24} \quad [(C) \text{ is Correct}]$$



$$(D) P(E^C \cap F^C \cap G^C) = 1 - P(E \cup F \cup G) \geq 1 - \frac{13}{24}$$

$$\Rightarrow P(E^C \cap F^C \cap G^C) \geq \frac{11}{24} \quad [(D) \text{ is Incorrect}]$$

$$(A) P(E) = \frac{1}{8} \geq P(E \cap F \cap G^C) + P(E \cap F \cap G)$$

$$\Rightarrow \frac{1}{8} \geq P(E \cap F \cap G^C) + \frac{1}{10} \Rightarrow \frac{1}{8} - \frac{1}{10} \geq P(E \cap F \cap G^C)$$

$$\Rightarrow \frac{1}{40} \geq P(E \cap F \cap G^C) \quad [(A) \text{ is Correct}]$$

$$(B) P(F) = \frac{1}{6} \geq P(E^C \cap F \cap G) + P(E \cap F \cap G)$$

$$\Rightarrow \frac{1}{6} - \frac{1}{10} \geq P(E^C \cap F \cap G) \Rightarrow \frac{4}{60} \geq P(E^C \cap F \cap G)$$

$$G \Rightarrow \frac{1}{15} \geq P(E^C \cap F \cap G) \quad [(B) \text{ is Correct}]$$

8. Ans (A,B,C)

$$|I - EF| \neq 0; G = (I - EF)^{-1} \Rightarrow G^{-1} = I - EF$$

$$\text{Now, } G \cdot G^{-1} = I = G^{-1} \cdot G$$

$$\Rightarrow G(I - EF) = I = (I - EF)G$$

$$\Rightarrow G - GEF = I = G - EFG$$

$$\Rightarrow GEF = EFG \quad [C \text{ is Correct}]$$

$$(I - FE)(I + FGE)$$

$$= I + FGE - FE - FEFG$$

$$= I + FGE - FE - F(G - I)E$$

$$= I + FGE - FE - FGE + FE$$

$$= I \quad [(B) \text{ is Correct}]$$

$$(\text{So 'D' is Incorrect})$$

$$\text{We have}$$

$$(I - FE)(I + FGE) = I \quad \dots (I)$$

$$\text{Now}$$

$$FE(I + FGE)$$

$$= FE + FEFG = FE + F(G - I)E$$

$$= FE + FGE - FE = FGE$$

$$\Rightarrow |FE| |I + FGE| = |FGE| \Rightarrow |FE| \times \text{ from (1)}$$

$$\Rightarrow |FE| = |I - FE| |FGE| \quad (\text{option (A) is correct})$$

9. Ans (A,B)

$$S_n(x) = \sum_{k=1}^n \tan^{-1} \left(\frac{x}{1+kx(kx+x)} \right)$$

$$= \sum_{k=1}^n \tan^{-1} \left(\frac{(kx+x) - (kx)}{1+(kx+x)(kx)} \right)$$

$$S_n(x) = \tan^{-1}(nx+x) - \tan^{-1}x = \tan^{-1} \left(\frac{nx}{1+(n+1)x^2} \right)$$

$$(A) S_{10}(x) = \tan^{-1} \frac{10x}{1+11x^2} = \frac{\pi}{2} - \tan^{-1} \left(\frac{1+11x^2}{10x} \right)$$

($x > 0$)

$$(B) \lim_{n \rightarrow \infty} \cot(S_n(x)) = \lim_{n \rightarrow \infty} \frac{\frac{1}{n} + (1 + \frac{1}{n})x^2}{x} = x \quad (x > 0)$$

$$(C) S_3(x) = \tan^{-1} \frac{3x}{1+4x^2} = \frac{\pi}{4} \Rightarrow 4x^2 - 3x + 1 = 0$$

$\Rightarrow x \notin \mathbb{R}$

$$(D) \tan(S_n(x)) = \frac{nx}{1+(n+1)x^2}; \forall n \geq 1; x > 0$$

We need to check the validity of

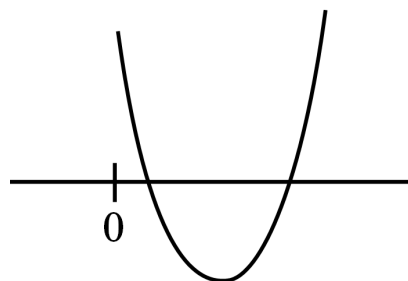
$$\frac{nx}{1+(n+1)x^2} \leq \frac{1}{2} \quad \forall n \geq 1; x > 0; n \in \mathbb{N}$$

$$\Rightarrow 2nx \leq (n+1)x^2 + 1$$

$$\Rightarrow (n+1)x^2 - 2nx + 1 \geq 0 \quad \forall n \geq 1; x > 0; n \in \mathbb{N}$$

Discriminant of $y = (n+1)x^2 - 2nx + 1$ is

$$D = 4n^2 - 4(n+1) \text{ and } n \in \mathbb{N}$$



$D < 0$ for $n = 1$; true for $x > 0$

$D > 0$ for $n \geq 2 \Rightarrow \exists$ some $x > 0$

for which $y < 0$ as both roots of $y = 0$ will be positive.

$$y = (n+1)x^2 - 2nx + 1, n \geq 2$$

So, $y \geq 0 \quad \forall n \geq 1; \forall x > 0; n \in \mathbb{N}$ is false.

10. Ans (B,D)

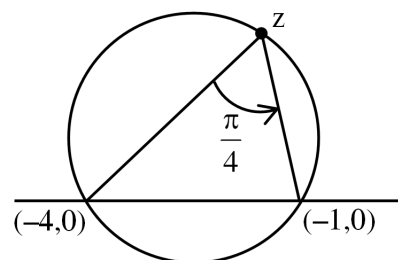
$$\arg \left(\frac{z+\alpha}{z+\beta} \right) = \frac{\pi}{4} \text{ implies } z \text{ is}$$

on arc and $(-\alpha, 0)$ & $(-\beta, 0)$ subtend $\frac{\pi}{4}$ on z .

And z lies on $x^2 + y^2 + 5x - 3y + 4 = 0$

So put $y = 0$;

$$x^2 + 5x + 4 = 0 \Rightarrow x = -1; x = -4$$



$$\text{Now, } \arg \left(\frac{z+\alpha}{z+\beta} \right) = \frac{\pi}{4}$$

$$\Rightarrow z + \alpha = (z + \beta) \cdot e^{i\frac{\pi}{4}}$$

$$\text{So, } z + \beta = z + 4 \Rightarrow \beta = 4 \text{ \& } z + \alpha = z + 1 \Rightarrow \alpha = 1$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (76.25)

p_1 = probability that maximum of chosen numbers is at least 81

$p_1 = 1$ - probability that maximum of chosen number is at most 80

$$p_1 = 1 - \frac{80 \times 80 \times 80}{100 \times 100 \times 100} = 1 - \frac{64}{125}$$

$$p_1 = \frac{61}{125}$$

$$\frac{625p_1}{4} = \frac{625}{4} \times \frac{61}{125} = \frac{305}{4} = 76.25$$

the value of $\frac{625p_1}{4}$ is 76.25

2. Ans (24.50)

p_2 = probability that minimum of chosen numbers is at most 40

= 1 - probability that minimum of chosen numbers is at least 41

=

$$1 - \left(\frac{600}{100} \right)^3$$

$$= 1 - \frac{27}{125} = \frac{98}{125}$$

$$\therefore \frac{125}{4} p_2 = \frac{125}{4} \times \frac{98}{125} = 24.50$$

3. Ans (1.00)

$$7x + 8y + 9z - (\gamma - 1) = A(4x + 5y + 6z - \beta) + B(x + 2y + 3z - \alpha)$$

$$x : 7 = 4A + B$$

$$y : 8 = 5A + 2B$$

$$A = 2, B = -1$$

$$\text{const. term : } -(\gamma - 1) = -A\beta - \alpha B \Rightarrow -(\gamma - 1) \equiv 2\beta + \alpha$$

$$\alpha - 2\beta + \gamma = 1$$

$$M = \begin{pmatrix} \alpha & 2 & \gamma \\ \beta & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} = \alpha - 2\beta + \gamma = 1$$

$$\text{Plane P : } x - 2y + z = 1$$

$$\text{Perpendicular distance}$$

$$= \left| \frac{3}{\sqrt{6}} \right| = P \Rightarrow D = P^2 = \frac{9}{6} = 1.5$$

4. Ans (1.50)

$$7x + 8y + 9z - (\gamma - 1) = A(4x + 5y + 6z - \beta) + B(x + 2y + 3z - \alpha)$$

$$x : 7 = 4A + B$$

$$y : 8 = 5A + 2B$$

$$A = 2, B = -1$$

$$\text{const. term : } -(\gamma - 1) = -A\beta - \alpha B \Rightarrow -(\gamma - 1) \equiv 2\beta + \alpha$$

$$\alpha - 2\beta + \gamma = 1$$

$$M = \begin{pmatrix} \alpha & 2 & \gamma \\ \beta & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} = \alpha - 2\beta + \gamma = 1$$

$$\text{Plane P : } x - 2y + z = 1$$

$$\text{Perpendicular distance}$$

$$= \left| \frac{3}{\sqrt{6}} \right| = P \Rightarrow D = P^2 = \frac{9}{6} = 1.5$$

5. Ans (9.00)

$$P(x, y) \left| \frac{\sqrt{2}x + y - 1}{\sqrt{3}} \right| \left| \frac{\sqrt{2}x - y + 1}{\sqrt{3}} \right| = \lambda^2$$

$$\left| \frac{2x^2 - (y-1)^2}{3} \right| = \lambda^2$$

$$C : |2x^2 - (y-1)^2| = 3\lambda^2$$

$$\text{line } y = 2x + 1, RS = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2},$$

$$R(x_1, y_1) \text{ and } S(x_2, y_2)$$

$$y_1 = 2x_1 + 1 \text{ and } y_2 = 2x_2 + 1 \Rightarrow (y_1 - y_2) = 2(x_1 - x_2)$$

$$RS = \sqrt{5(x_1 - x_2)^2} = \sqrt{5} |x_1 - x_2|$$

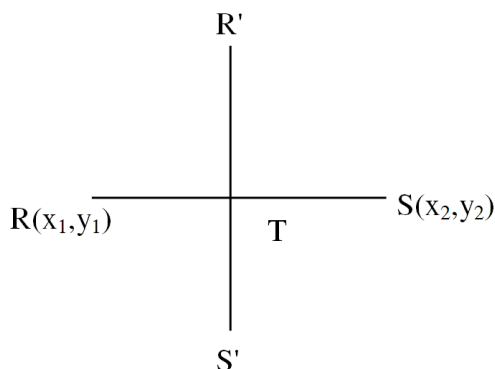
$$\text{solve curve C and line } y = 2x + 1 \text{ we get}$$

$$|2x^2 - (2x)^2| = 3\lambda^2 \Rightarrow x^2 = \frac{3\lambda^2}{2}$$

$$RS = \sqrt{5} \left| \frac{2\sqrt{3}\lambda}{\sqrt{2}} \right| = \sqrt{30}\lambda = \sqrt{270}$$

$$\Rightarrow 30\lambda^2 = 270 \Rightarrow \lambda^2 = 9$$

6. Ans (77.14)



$$\perp \text{ bisector of RS } T \equiv \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$\text{Here } x_1 + x_2 = 0$$

$$T = (0, 1)$$

$$\text{Equation of}$$

$$R'S' : (y - 1) = -\frac{1}{2}(x - 0) \Rightarrow x + 2y = 2$$

$$R'(a_1, b_1) S'(a_2, b_2)$$

$$D = (a_1 - a_2)^2 + (b_1 - b_2)^2 = 5(b_1 - b_2)^2$$

$$\text{solve } x + 2y = 2 \text{ and } |2x^2 - (y-1)^2| = 3\lambda^2$$

$$|8(y-1)^2 - (y-1)^2| = 3\lambda^2 \Rightarrow (y-1)^2 = \left(\frac{\sqrt{3}\lambda}{\sqrt{7}} \right)^2$$

$$y - 1 = \pm \frac{\sqrt{3}\lambda}{\sqrt{7}} \Rightarrow b_1 = 1 + \frac{\sqrt{3}\lambda}{\sqrt{7}}, b_2 = 1 - \frac{\sqrt{3}\lambda}{\sqrt{7}}$$

$$D = 5 \left(\frac{2\sqrt{3}\lambda}{\sqrt{7}} \right)^2 = \frac{5 \times 4 \times 3\lambda^2}{7} = \frac{5 \times 4 \times 27}{7} = 77.14$$

PART-3 : MATHEMATICS

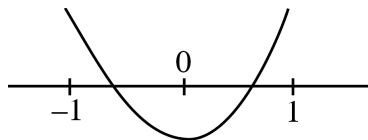
SECTION-II (ii)

7. **Ans (4)**

$$3x^2 + x - 1 = 4|x^2 - 1|$$

$$\text{If } x \in [-1, 1],$$

$$3x^2 + x - 1 = -4x^2 + 4 \Rightarrow 7x^2 + x - 5 = 0$$

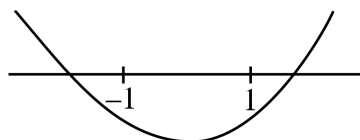


$$\text{say } f(x) = 7x^2 + x - 5$$

$$f(1) = 3; f(-1) = 1; f(0) = -1$$

[Two Roots]

$$\text{If } x \in (-\infty, -1] \cup [1, \infty)$$



$$3x^2 + x - 1 = 4x^2 - 4 \Rightarrow x^2 - x - 3 = 0$$

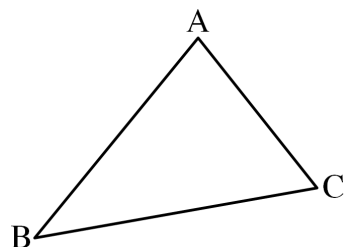
$$\text{Say } g(x) = x^2 - x - 3$$

$$g(-1) = -1; g(1) = -3$$

[Two Roots]

So total 4 roots.

8. **Ans (2)**



$$\text{Given } c = \sqrt{23}; a = 3; b = 4$$

$$\cot A = \frac{\cos A}{\sin A} = \frac{b^2 + c^2 - a^2}{2bc \sin A}$$

$$= \frac{b^2 + c^2 - a^2}{2.2\Delta} \left\{ \Delta = \frac{1}{2}bc \sin A \right\}$$

$$\cot A = \frac{b^2 + c^2 - a^2}{4\Delta}$$

$$\text{Similarly, } \cot B = \frac{a^2 + c^2 - b^2}{4\Delta} \quad \&$$

$$\cot C = \frac{a^2 + b^2 - c^2}{4\Delta}$$

$$\therefore \frac{\cot A + \cot C}{\cot B} = \frac{b^2 + c^2 - a^2 + a^2 + b^2 - c^2}{a^2 + c^2 - b^2}$$

$$= \frac{2b^2}{a^2 + c^2 - b^2} = \frac{32}{16} = 2$$

9. **Ans (7)**

$$\text{Given, } |\vec{u}| = 1; |\vec{v}| = 1; \vec{u} \cdot \vec{v} \neq 0; \vec{u} \cdot \vec{w} = 1;$$

$$\vec{v} \cdot \vec{w} = 1;$$

$$\vec{w} \cdot \vec{w} = |\vec{w}|^2 = 4 \Rightarrow |\vec{w}| = 2;$$

$$[\vec{u} \ \vec{v} \ \vec{w}] = \sqrt{2}$$

$$\text{and } [\vec{u} \ \vec{v} \ \vec{w}]^2 = \begin{vmatrix} \vec{u} \cdot \vec{u} & \vec{u} \cdot \vec{v} & \vec{u} \cdot \vec{w} \\ \vec{v} \cdot \vec{u} & \vec{v} \cdot \vec{v} & \vec{v} \cdot \vec{w} \\ \vec{w} \cdot \vec{u} & \vec{w} \cdot \vec{v} & \vec{w} \cdot \vec{w} \end{vmatrix} = 2$$

$$\Rightarrow \begin{vmatrix} 1 & \vec{u} \cdot \vec{v} & 1 \\ \vec{u} \cdot \vec{v} & 1 & 1 \\ 1 & 1 & 4 \end{vmatrix} = 2$$

$$\Rightarrow \vec{u} \cdot \vec{v} = \frac{1}{2}$$

$$\text{So, } |3\vec{u} + 5\vec{v}| = \sqrt{9|\vec{u}|^2 + 25|\vec{v}|^2 + 2 \cdot 3 \cdot 5 \vec{u} \cdot \vec{v}}$$

$$= \sqrt{9 + 25 + 30 \left(\frac{1}{2} \right)} = \sqrt{49} = 7$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

FULL SYLLABUS

16-02-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-I)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B,D	A,D	B,C	B,D	A,C,D	A,B
SECTION-I (ii)	Q.	7	8	9	10		
	A.	A	C	A	B		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	0.30	10.00	0.18	0.16	100.00	60.00
SECTION-II (ii)	Q.	7	8	9			
	A.	3	9	6			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B	C,D	B,C,D	A,B,C	A,B,D	A,B,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	A	D	C	D		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	0.21 or 0.22	0.86	3.57	1.23	1.87 or 1.88	18.75
SECTION-II (ii)	Q.	7	8	9			
	A.	10	30	6			

PART-3 : MATHEMATICS

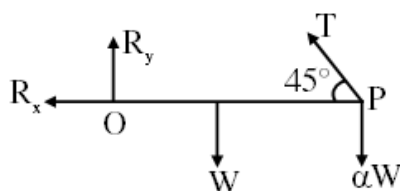
SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B,D	A,B	A,B,C	A,C	A,B,C	A,B,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	D	B	C	D		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	1.50	2.00	57.00	6.00	2.00	1.50
SECTION-II (ii)	Q.	7	8	9			
	A.	214	4	182			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,B,D)



$$R_y + \frac{T}{\sqrt{2}} = W + \alpha W \quad \dots(i)$$

$$R_x = \frac{T}{\sqrt{2}} \quad \dots(ii)$$

Taking torque about 'O'

$$W \frac{l}{2} + \alpha W l = \frac{T}{\sqrt{2}} l$$

$$T = \sqrt{2} \left(\frac{W}{2} + \alpha W \right) \quad \dots(iii)$$

$$R_x = \frac{T}{\sqrt{2}} = \left(\frac{W}{2} + \alpha W \right)$$

Taking torque about P

$$R_y l = W \frac{l}{2}$$

$$R_y = \frac{W}{2};$$

when $T = T_{\max}$

$$2\sqrt{2}W = \sqrt{2} \left(\frac{W}{2} + \alpha W \right)$$

$$\text{we get } \alpha = \frac{3}{2}$$

2. **Ans (A,D)**

$$f = f_s \left(\frac{v}{v-u} \right)$$

$$(A) f = f_0 \left(\frac{v}{v-0.8v} \right) = 5f_0$$

$$(B) f = 2f_0 \left(\frac{v}{v-0.8v} \right) = 10f_0$$

$$(C) f = 0.5f_0 \left(\frac{v}{v-0.8v} \right) = 2.5f_0$$

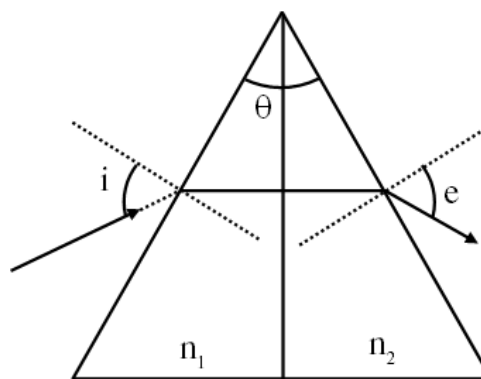
$$(D) f = 1.5f_0 \left(\frac{v}{v-0.5v} \right) = 3f_0$$

close

All odd harmonics are available in closed pipe

therefore correct Ans (A,D)

3. **Ans (B,C)**



$$1 \times \sin i = \mu \sin \left(\frac{\theta}{2} \right)$$

$$\sin i = \frac{3}{4}$$

$$n_1 \sin 30^\circ = 1 \sin(e)$$

on differentiating both sides

$$dn \sin 30^\circ = de \cos(e)$$

$$de = \frac{dn}{2 \cos(e)}$$

$$= \frac{dn}{2 \times \frac{\sqrt{7}}{4}}$$

$$de = \frac{2}{\sqrt{7}} dn \Rightarrow de < dn$$

$$de = \frac{2.8 \times 10^{-3} \times 2}{\sqrt{7}} = 2.11 \text{ mrad}$$

4. **Ans (B,D)**

$$\vec{S} = [\vec{E} \times \vec{B}] \frac{1}{\mu_0}$$

$\vec{S}$ is pointing vector denotes flow of energy per unit area per unit time

$$\vec{S} = \frac{\text{watt}}{\text{m}^2}$$

Hence B, D are correct

5. Ans (A,C,D)

$$N \rightarrow P + Q$$

$$\text{Energy released} = (m_N - m_P - m_Q)c^2 = \delta c^2$$

This will be distributed kinetic energy of P and Q

$$\Rightarrow E_P + E_Q = \delta c^2 \quad \dots(i)$$

By conservation of momentum

$$v_P = \frac{p}{m_P} \quad \leftarrow \text{P} \quad \text{Q} \rightarrow \frac{p}{m_Q} = v_Q$$

$$\text{So } \frac{v_P}{v_Q} = \frac{M_Q}{M_P} \quad \dots(ii)$$

$$\text{Kinetic energy be written as } KE = \frac{p^2}{2m}$$

Hence divided in inverse ratio of masses.

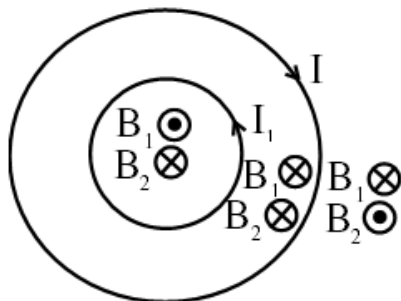
$$E_P = \frac{M_Q}{M_P + M_Q} c^2 \delta \quad \dots(iii)$$

$$\text{By equation (i)} \Rightarrow \frac{p^2}{2M_P} + \frac{p^2}{2M_Q} = \delta c^2$$

$$\Rightarrow \frac{p^2}{2\mu} = \delta c^2 \Rightarrow p = c\sqrt{2\mu\delta}$$

Ans. (A,C,D)

6. Ans (A,B)



$$(A) d\vec{B} = \frac{\mu_0 I d\vec{\ell} \times \vec{r}}{4\pi r^3}$$

$\vec{d\ell}$ is in xy plane & $\vec{r}$ is also in xy plane

so $d\vec{B}$ is perpendicular to xy plane

(B) Due to symmetry it depends only on

$$r = \sqrt{x^2 + y^2}$$

$$(C) \text{ At centre } B_1 = \frac{\mu_0 I_1}{2R}; B_2 = \frac{\mu_0 I_2}{4R} \Rightarrow B_2 > B_1$$

but as we approach towards first loop B_1 increases to infinity hence B_1 dominates.

So it would be zero at some point between inner loops and centre.

Ans. (A,B)

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (A)

$$\phi = Li = \frac{\mu_0 m}{2\pi r^3} \times \pi a^2$$

$$\Rightarrow i = \frac{\mu_0 m \pi a^2}{2\pi r^3 L}$$

$$\Rightarrow i \propto \frac{m}{r^3}$$

$$m' = \pi a^2 i = \frac{\mu_0 m \pi^2 a^4}{2\pi r^3 L}$$

$$F = \frac{k m^2 \pi^2 a^4}{2\pi r^7 L}$$

$$W = \int F dr \propto \int \frac{m^2 dr}{r^7}$$

$$W \propto \frac{m^2}{r^6}$$

8. Ans (C)

$$\phi = Li = \frac{\mu_0 m}{2\pi r^3} \times \pi a^2$$

$$\Rightarrow i = \frac{\mu_0 m \pi a^2}{2\pi r^3 L}$$

$$\Rightarrow i \propto \frac{m}{r^3}$$

$$m' = \pi a^2 i = \frac{\mu_0 m \pi^2 a^4}{2\pi r^3 L}$$

$$F = \frac{k m^2 \pi^2 a^4}{2\pi r^7 L}$$

$$W = \int F dr \propto \int \frac{m^2 dr}{r^7}$$

$$W \propto \frac{m^2}{r^6}$$

9. Ans (A)

$$\text{Finally } V_L = \frac{3V_0}{2}, V_R = \frac{V_0}{2}$$

$$C_V = \frac{R}{\gamma - 1} = 2R \Rightarrow \gamma - 1 = \frac{1}{2}$$

$$\gamma = \frac{3}{2}$$

$$T_0 V_0^{\gamma-1} = T_R \left(\frac{V_0}{2} \right)^{\gamma-1}$$

$$\frac{T_R}{T_0} = \sqrt{2}$$

$$\rho \left(\frac{V_0}{2} \right)^{\gamma} = P_0 V_0^{\gamma} \Rightarrow P = P_0 \times 2^{\frac{3}{2}}$$

$$\frac{PV}{T_L} = \frac{P_0 V_0}{T_0} \Rightarrow T_L = 2^{\frac{3}{2}} \times \frac{3}{2} T_0 = 3\sqrt{2} T_0$$

$$Q = nC_V \Delta T_1 + nC_V \Delta T_2$$

$$= 1 \times 2R \times (3\sqrt{2} - 1) T_0 + 1 \times 2R \times (\sqrt{2} - 1) T_0$$

$$\frac{Q}{RT_0} = 2(3\sqrt{2} - 1) + 2(\sqrt{2} - 1)$$

$$= 8\sqrt{2} - 4$$

10. Ans (B)

$$\text{Finally } V_L = \frac{3V_0}{2}, V_R = \frac{V_0}{2}$$

$$C_V = \frac{R}{\gamma - 1} = 2R \Rightarrow \gamma - 1 = \frac{1}{2}$$

$$\gamma = \frac{3}{2}$$

$$T_0 V_0^{\gamma-1} = T_R \left(\frac{V_0}{2} \right)^{\gamma-1}$$

$$\frac{T_R}{T_0} = \sqrt{2}$$

$$\rho \left(\frac{V_0}{2} \right)^{\gamma} = P_0 V_0^{\gamma} \Rightarrow P = P_0 \times 2^{\frac{3}{2}}$$

$$\frac{PV}{T_L} = \frac{P_0 V_0}{T_0} \Rightarrow T_L = 2^{\frac{3}{2}} \times \frac{3}{2} T_0 = 3\sqrt{2} T_0$$

$$Q = nC_V \Delta T_1 + nC_V \Delta T_2$$

$$= 1 \times 2R \times (3\sqrt{2} - 1) T_0 + 1 \times 2R \times (\sqrt{2} - 1) T_0$$

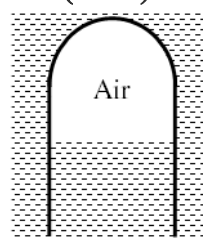
$$\frac{Q}{RT_0} = 2(3\sqrt{2} - 1) + 2(\sqrt{2} - 1)$$

$$= 8\sqrt{2} - 4$$

PART-1 : PHYSICS

SECTION-II (i)

1. Ans (0.30)



When it starts sinking

$$F_B = mg$$

$$\rho_0 (V_{\text{glass}} + V_{\text{gas}}) = m$$

$$1(2 + V_{\text{gas}}) = 5 \Rightarrow V_{\text{gas}} = 3 \text{ cc}$$

Hence $\Delta V = 0.3 \text{ cc}$.

2. Ans (10.00)

Isothermal process for air

$$P_1 V_1 = P_2 V_2$$

$$10^5 (3.3) = P_2 (3)$$

$$P_2 = 1.1 \times 10^5$$

$$\Delta P = P_2 - P_1 = 1.1 \times 10^5 - 10^5$$

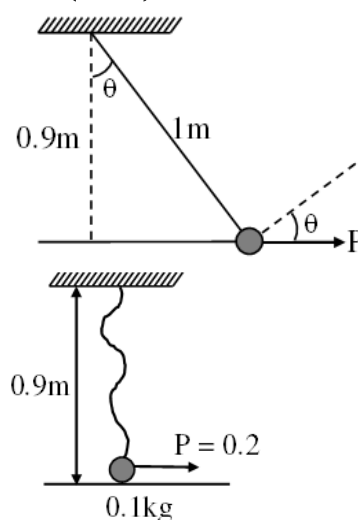
$$= 0.1 \times 10^5$$

$$= 10 \times 10^3 \text{ Pascal}$$

$$= Y \times 10^3 \text{ Pascal}$$

So $Y = 10$

3. Ans (0.18)



$$L = P \times 0.9 = 0.18 \text{ kgm}^2/\text{s}$$

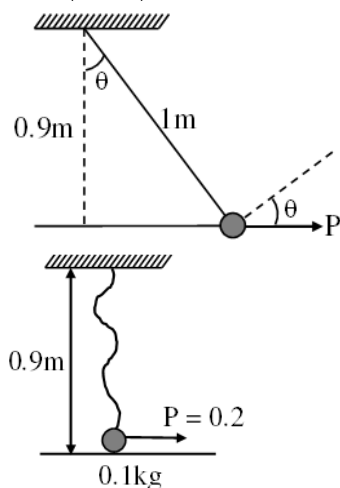
Just after string becomes taut; there will be no velocity along the string.

$$\therefore V_{\perp} = \frac{P \cos \theta}{m} = \frac{0.2 \times 0.9}{1 \times 0.1} = 1.8 \text{ m/s}$$

$$\therefore K = \frac{1}{2} m v_{\perp}^2 = \frac{1}{2} \times 0.1 \times 1.8^2$$

$$= 0.162 \text{ J}$$

4. Ans (0.16)



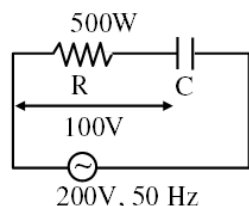
$$L = P \times 0.9 = 0.18 \text{ kgm}^2/\text{s}$$

Just after string becomes taut; there will be no velocity along the string.

$$\therefore V_{\perp} = \frac{P \cos \theta}{m} = \frac{0.2 \times 0.9}{1 \times 0.1} = 1.8 \text{ m/s}$$

$$\therefore K = \frac{1}{2} m v_{\perp}^2 = \frac{1}{2} \times 0.1 \times 1.8^2 = 0.162 \text{ J}$$

5. Ans (100.00)



$$\sqrt{V_C^2 + V_R^2} = \epsilon_{\text{rms}}$$

$$\Rightarrow V_C^2 + 100^2 = 200^2$$

$$V_C = 100\sqrt{3} \text{ V} \dots(i)$$

$$\tan \phi = \frac{V_C}{V_R} = \frac{100\sqrt{3}}{100}$$

$$\therefore \phi = 60^\circ \dots(ii)$$

$$P = I_{\text{rms}} \epsilon_{\text{rms}} \cos \phi$$

$$= \frac{\epsilon_{\text{rms}}^2}{Z} \frac{1}{2}$$

$$500 = \frac{200}{Z} \frac{1}{2}$$

$$\therefore Z = 40 \Omega \dots(iii)$$

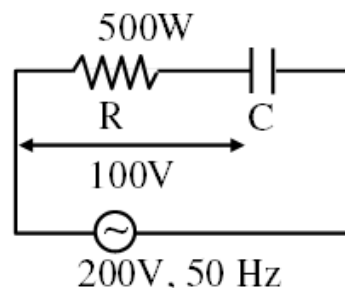
$$\cos \phi = \frac{R}{Z} \Rightarrow \frac{1}{2} = \frac{R}{40}$$

$$\therefore R = 20$$

$$\& x_C = \sqrt{Z^2 - R^2} = \sqrt{40^2 - 20^2} = 20\sqrt{3} \Omega$$

$$\Rightarrow \frac{1}{\omega C} = 20\sqrt{3} \therefore C = 100$$

6. Ans (60.00)



$$\sqrt{V_C^2 + V_R^2} = \epsilon_{\text{rms}}$$

$$\Rightarrow V_C^2 + 100^2 = 200^2$$

$$V_C = 100\sqrt{3} \text{ V} \dots(i)$$

$$\tan \phi = \frac{V_C}{V_R} = \frac{100\sqrt{3}}{100}$$

$$\therefore \phi = 60^\circ \dots(ii)$$

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$$= \frac{\epsilon_{\text{rms}}^2}{Z} \frac{1}{2}$$

$$500 = \frac{200}{Z} \frac{1}{2}$$

$$\therefore Z = 40 \Omega \dots(iii)$$

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$$\therefore R = 20$$

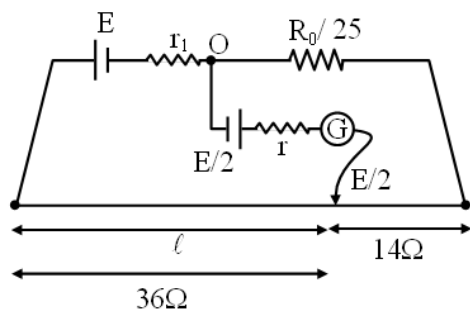
$$\& x_C = \sqrt{Z^2 - R^2} = \sqrt{40^2 - 20^2} = 20\sqrt{3} \Omega$$

$$\Rightarrow \frac{1}{\omega C} = 20\sqrt{3} \therefore C = 100$$

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (3)



$$i \left(\frac{R_0}{2} + 0.28R_0 \right) = \frac{E_0}{2}$$

$$i \times 0.78R_0 = \frac{E_0}{2}$$

$$i = \frac{E_0}{2 \times 0.78R_0} = \frac{E_0}{r_1 + \frac{3}{2}R_0}$$

$$r_1 + 1.5R_0 = 1.56R_0$$

$$r_1 = 0.06R_0$$

$$= 0.06 \times 50 = 3 \Omega$$

8. Ans (9)

Circular orbits

$$T = 2\pi \sqrt{\frac{R^2}{GM_S}}$$

Binary stars

$$nT = 2\pi \sqrt{\frac{(9R)^3}{G(3M_S + 6M_S)}}$$

$$n \times 2\pi \sqrt{\frac{R^3}{GM_S}} = 9 \times 2\pi \sqrt{\frac{R^3}{GM_S}}$$

$$n = 9$$

9. Ans (6)

For P & Q

$$E_1 - 4 = E_P$$

$$E_1 - 4.5 = E_Q$$

$$E_P = 2E_Q$$

$$E_1 - 4 = 2(E_1 - 4.5)$$

$$E_1 = 5 \text{ eV}$$

$$E_P = 1 \text{ eV}, E_Q = E_R = 0.5 \text{ eV}$$

$$\text{For } E_2 - 5.5 = 0.5$$

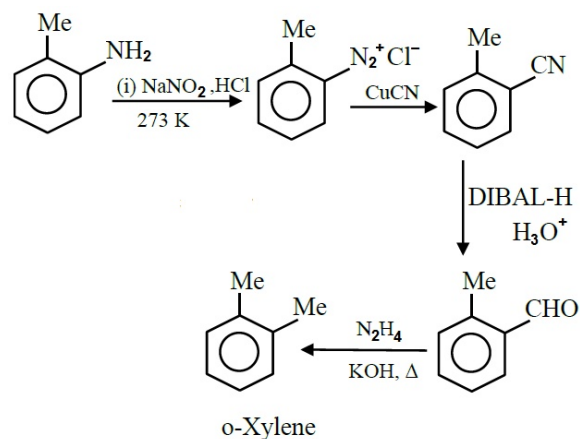
$$E_2 = 6 \text{ eV}$$

PART-2 : CHEMISTRY

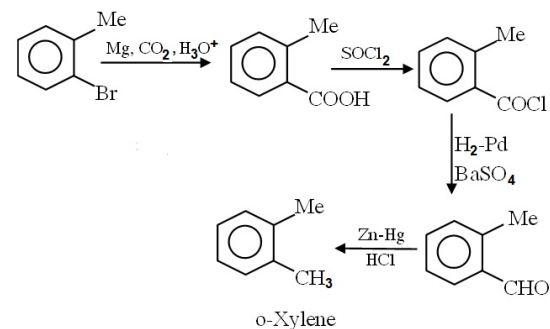
SECTION-I (i)

1. Ans (A, B)

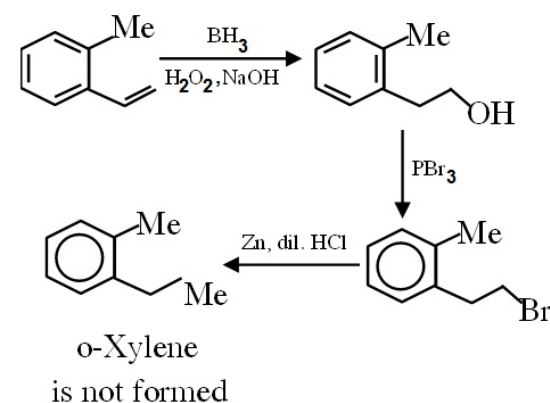
(A)



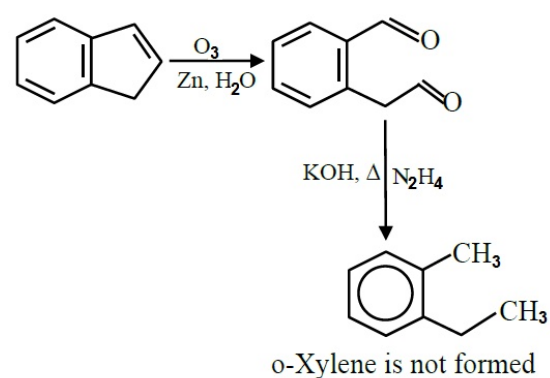
(B)



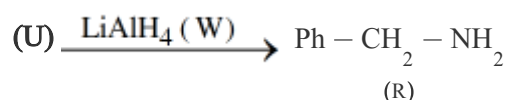
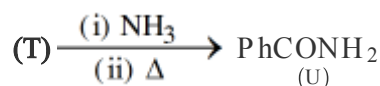
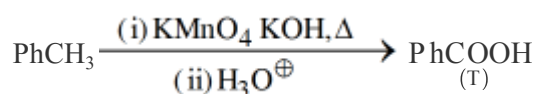
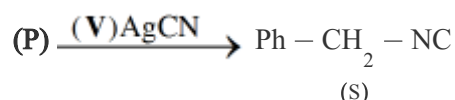
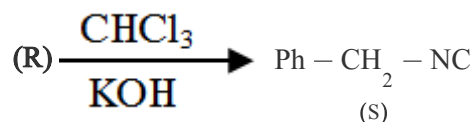
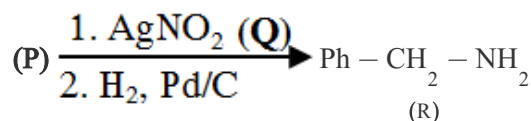
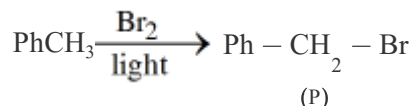
(C)



(D)



2. Ans (C,D)



3. Ans (B,C,D)

$$\frac{dp}{dt} = k[x]^1$$

$$2x + y \rightarrow p$$

$$t = 0 \quad 2 \quad 1$$

$$t = 50 \text{ s} \quad (2-1) \quad (1-0.5) \quad 0.5$$

$$-\frac{1}{2} \frac{dx}{dt} = \frac{dp}{dt} = k[x]^1$$

$$-\frac{dx}{dt} = 2k[x]^1$$

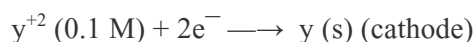
$$2k = \frac{\ln 2}{50} \Rightarrow k = \frac{\ln 2}{100}$$

$$\text{At } 50 \text{ sec } -\frac{dx}{dt} = 2k \times (1)^1 = \frac{\ln 2}{50}$$

$$\text{At } 100 \text{ sec } -\frac{1}{2} \frac{dx}{dt} = \frac{-dy}{dt}$$

$$\Rightarrow -\frac{dy}{dt} = \frac{\ln 2}{100} \times \frac{1}{2} \left\{ \frac{-dy}{dt} = k[x]^1 \right\}$$

4. Ans (A,B,C)



$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.06}{2} \log \frac{x^{+2}}{y^{+2}}$$

$$\log \frac{x^{+2}}{y^{+2}}$$

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} + 0.06$$

$$(\text{A}) \text{ Cd and Ni } E_{\text{cell}}^{\circ} = +0.4 - 0.24 ; E_{\text{cell}} = 0.22$$

$$(\text{B}) \text{ Cd and Fe } E_{\text{cell}}^{\circ} = -0.04 ; E_{\text{cell}} = 0.02$$

$$(\text{C}) \text{ Ni and Pb } E_{\text{cell}}^{\circ} = 0.11 ; E_{\text{cell}} = 0.17$$

$$(\text{D}) \text{ Ni and Fe } E_{\text{cell}}^{\circ} = -0.2 ; E_{\text{cell}} = -0.14$$

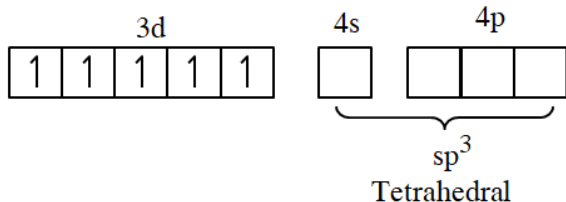
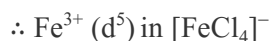
since in (A) (B) (C) E_{cell} is positive hence answer is (A) (B) (C).

5. Ans (A,B,D)

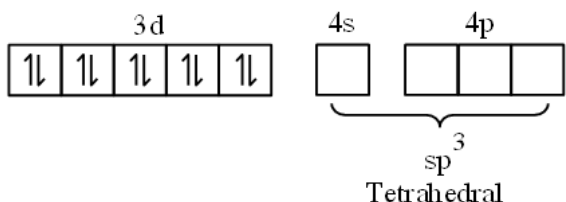
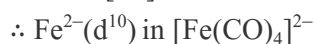
(A) $[\text{FeCl}_4]^-$



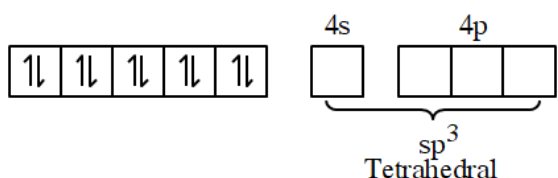
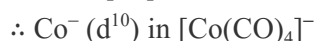
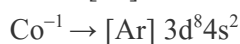
Cl^- is W.F.L. and does not pair up the unpaired electron of central metal atom.



$[\text{Fe}(\text{CO})_4]^{2-}$



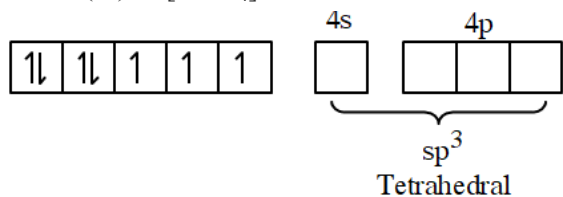
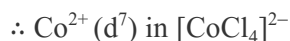
(B) $[\text{Co}(\text{CO})_4]^-$



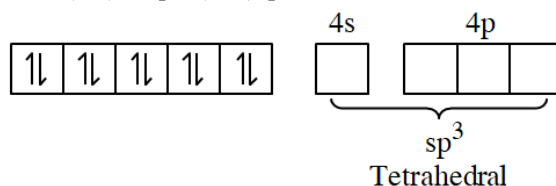
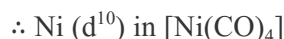
$[\text{CoCl}_4]^{2-}$



Cl^- is W.F.L. and does not pair up the unpaired electron of central metal atom.



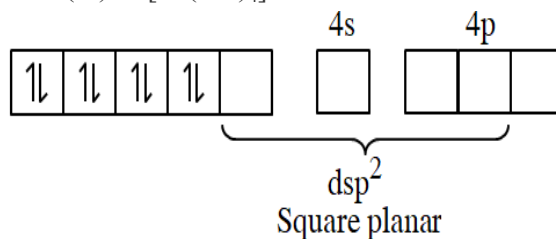
(C) $[\text{Ni}(\text{CO})_4]$



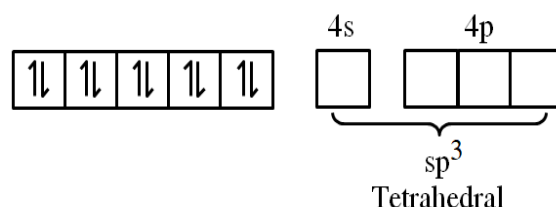
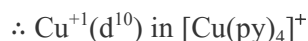
$[\text{Ni}(\text{CN})_4]^{2-}$



CN^- is S.F.L. and pair up the unpaired electron of central metal atom.



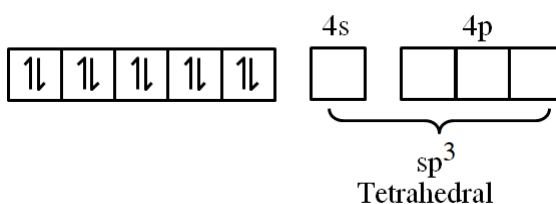
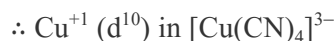
(D) $[\text{Cu}(\text{py})_4]^+$



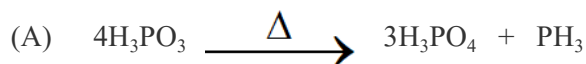
$[\text{Cu}(\text{CN})_4]^{3-}$



CN^- is S.F.L. and pair up the unpaired electron of central metal atom.



6. Ans (A,B,D)

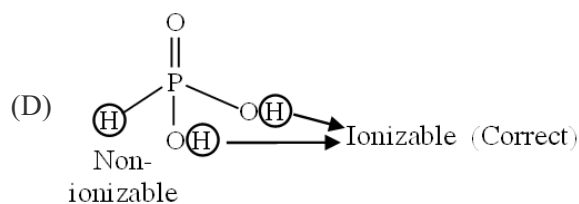
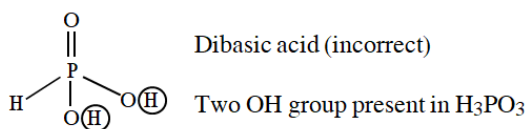


(correct)

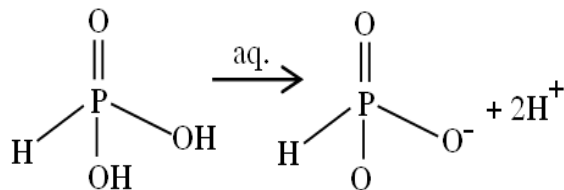
(B) H_3PO_4 has "P" in its highest oxidation state, hence cannot act as a reducing agent

(correct)

(C)



The hydrogen which is directly attached to phosphorous does not ionized in water.

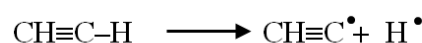
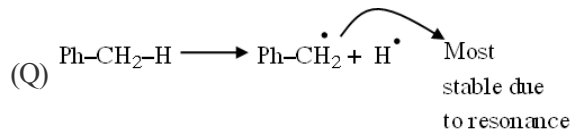
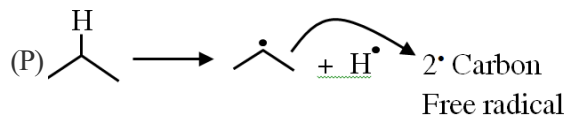


PART-2 : CHEMISTRY

SECTION-I (ii)

7. Ans (A)

Most stability of radical, less is the bond energy



(S)

More % s-Character
decreases stability of
free radical

Q require least BDE and S Required maximum BDE
Max BDE

So, Order of BDE $\text{Q} < \text{P} < \text{R} < \text{S}$

8. Ans (D)

Initiation step is endothermic hence option (A) is wrong.

Propagation step involving $\cdot\text{CH}_3$ formation is endothermic hence option (B) is wrong.

Propagation step involving CH_3Cl formation is exothermic hence option (C) is wrong.

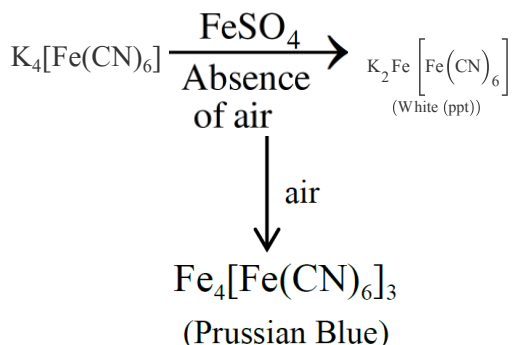
Reaction



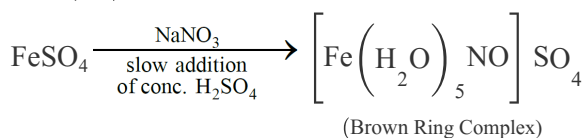
Overall reaction is exothermic with

$\Delta H^\circ = -25 \text{ KCal/mol}$, hence option (D) is correct.

9. Ans (C)



10. Ans (D)



PART-2 : CHEMISTRY

SECTION-II (i)

1. Ans (0.21 or 0.22)

$$K_a = \frac{\Lambda_m^2 C}{\Lambda_m^\circ (\Lambda_m^\circ - \Lambda_m)}$$

$$K_a =$$

$$\frac{(y \times 10^2)^2 \times C}{4 \times 10^2 (4 \times 10^2 - y \times 10^2)}$$

$$= \frac{(3y \times 10^2)^2 \times \frac{C}{20}}{4 \times 10^2 (4 \times 10^2 - 3y \times 10^2)}$$

$$\Rightarrow \frac{1}{(4-y)} = \frac{9}{20(4-3y)}$$

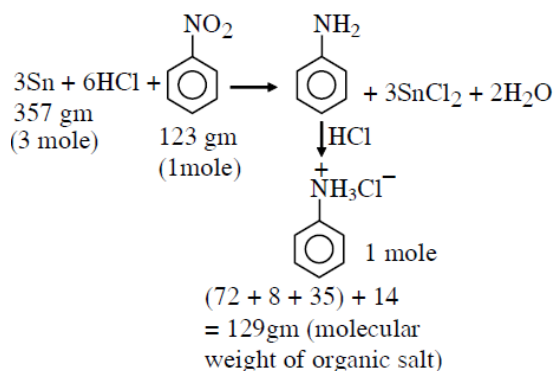
$$\Rightarrow y = \frac{44}{51}$$

$$\alpha = \frac{\frac{44}{51} \times 10}{4 \times 10^2} \quad \alpha = 0.2156 \quad (\alpha = 0.22 \text{ or } 0.21)$$

$$y = 0.86$$

3. Ans (3.57)

The value of x is



So to get 1.29 gm organic salt.

We have to form 0.01 mole salt.

So 0.01 mole nitrobenzene is required.

0.03 mole Sn is required.

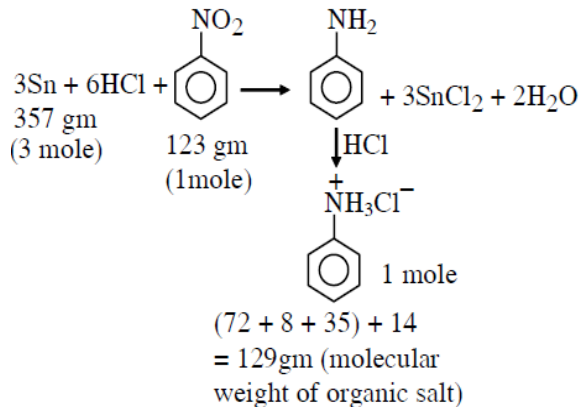
So the amount of nitrobenzene = 0.01 × 123 = 1.23 gm

the amount of Sn required = 0.01 × 357 = 3.57 gm

Ans. 3.57 & 1.23

4. Ans (1.23)

The value of y is



So to get 1.29 gm organic salt.

We have to form 0.01 mole salt.

So 0.01 mole nitrobenzene is required.

0.03 mole Sn is required.

So the amount of nitrobenzene = 0.01 × 123 = 1.23 gm

the amount of Sn required = 0.01 × 357 = 3.57 gm

Ans. 3.57 & 1.23

3. Ans (A,B,C)

(A) Let $g(x) = f(x) - 3 \cos 3x$

Now

$$\int_0^{\pi/3} g(x) dx = \int_0^{\pi/3} f(x) dx - 3 \int_0^{\pi/3} \cos 3x dx = 0$$

Hence $g(x) = 0$ has a root in $(0, \frac{\pi}{3})$

(B) Let $h(x) = f(x) - 3 \sin 3x + \frac{6}{\pi}$

Now

$$\int_0^{\pi/3} h(x) dx = \int_0^{\pi/3} f(x) dx - 3 \int_0^{\pi/3} \sin 3x dx + \int_0^{\pi/3} \frac{6}{\pi} dx$$

$$= 0 - 2 + 2 = 0$$

Hence $h(x) = 0$ has a root in $(0, \frac{\pi}{3})$

(C)

$$\lim_{x \rightarrow 0} \frac{x \int_0^x f(t) dt}{1 - e^{x^2}} = \lim_{x \rightarrow 0} \underbrace{\left(\frac{x^2}{1 - e^{x^2}} \right)}_{-1} \underbrace{\frac{\int_0^x f(t) dt}{x}}_{\text{Apply L' Hopital's Rule}}$$

$$= -1 \lim_{x \rightarrow 0} \frac{f(x)}{1} = -1$$

$$(\sin x) \int_0^x f(t) dt$$

$$(D) \lim_{x \rightarrow 0} \frac{(\sin x) \int_0^x f(t) dt}{x^2}$$

$$= \lim_{x \rightarrow 0} \underbrace{\left(\frac{\sin x}{x} \right)}_1 \underbrace{\frac{\int_0^x f(t) dt}{x}}_{\text{Apply L' Hopitals Rule}}$$

$$= 1 \lim_{x \rightarrow 0} \frac{f(x)}{1} = 1$$

4. Ans (A,C)

Integrating factor = $e^{\alpha x}$

$$\text{So } ye^{\alpha x} = \int xe^{(\alpha+\beta)x} dx$$

Case-I

$$\text{If } \alpha + \beta = 0 \quad ye^{\alpha x} = \frac{x^2}{2} + c$$

$$\text{It passes through } (1, 1) \Rightarrow C = e^{\alpha} - \frac{1}{2}$$

$$\text{So } ye^{\alpha x} = \frac{x^2 - 1}{2} + e^{\alpha}$$

for $\alpha = 1$

$$y = \frac{x^2}{2} e^{-x} + \left(e - \frac{1}{2} \right) e^{-x} \rightarrow (A)$$

Case-II

If $\alpha + \beta \neq 0$

$$ye^{\alpha x} = \frac{x \cdot e^{(\alpha+\beta)x}}{\alpha + \beta} - \frac{1}{\alpha + \beta} e^{(\alpha+\beta)x} dx$$

$$\Rightarrow ye^{\alpha x} = \frac{xe^{(\alpha+\beta)x}}{\alpha + \beta} - \frac{e^{(\alpha+\beta)x}}{(\alpha + \beta)^2} + c$$

$$\Rightarrow \text{So } c = e^{\alpha} - \frac{e^{\alpha+\beta}}{\alpha + \beta} + \frac{e^{\alpha+\beta}}{(\alpha + \beta)^2}$$

$$y = \frac{e^{\beta x}}{(\alpha + \beta)^2} ((\alpha + \beta)x - 1) + e^{-\alpha x} \left(e^x - \frac{e^{\alpha+\beta}}{\alpha + \beta} + \frac{e^{\alpha+\beta}}{(\alpha + \beta)^2} \right)$$

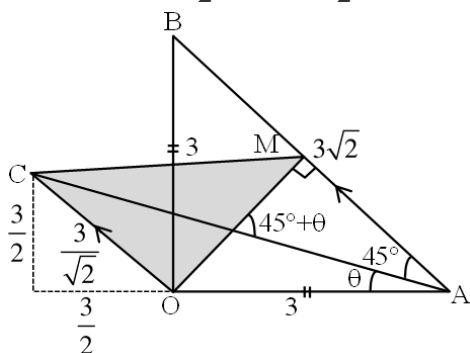
If $\alpha = \beta = 1$

$$y = \frac{e^x}{4} (2x - 1) + e^{-x} \left(e - \frac{e^2}{2} + \frac{e^2}{4} \right)$$

$$y = \frac{e^x}{2} \left(x - \frac{1}{2} \right) + e^{-x} \left(e - \frac{e^2}{4} \right) \rightarrow (c)$$

5. Ans (A,B,C)

$$\begin{aligned}\vec{OB} \times \vec{OC} &= \frac{1}{2} \vec{OB} \times (\vec{OB} - \lambda \vec{OA}) \\&= \frac{\lambda}{2} (\vec{OA} \times \vec{OB}) \\|\vec{OB}| \times |\vec{OC}| &= \frac{|\lambda|}{2} |\vec{OA}| \times |\vec{OB}| \quad (\text{Note } \vec{OA} \text{ \& } \vec{OB} \text{ are perpendicular}) \\ \Rightarrow \frac{9\lambda}{2} &= \frac{9}{2} \Rightarrow \lambda = 1 \quad (\text{given } \lambda > 0) \\ \text{So } \vec{OC} &= \frac{\vec{OB} - \vec{OA}}{2} = \frac{\vec{AB}}{2}\end{aligned}$$



M is mid point of AB

Note projection of $\vec{OC}$ on $\vec{OA} = -\frac{3}{2}$

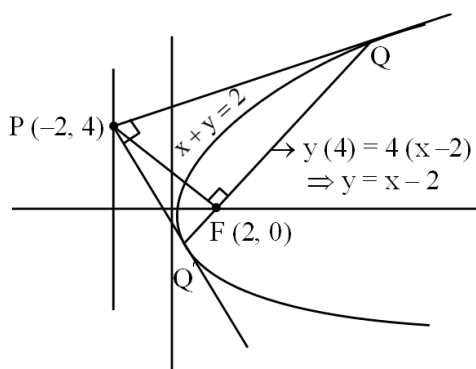
$$\tan \theta = \frac{1}{3}$$

$$\text{Area of } \triangle ABC = \frac{9}{2}$$

Acute angle between diagonals is

$$\tan^{-1} \left(\frac{1 + \frac{1}{3}}{1 - \frac{1}{3}} \right) = \tan^{-1} 2$$

6. Ans (A,B,D)



Note that P lies on directrix so triangle PQQ' is right angled, hence QQ' passes through focus F.

$$PF = 4\sqrt{2}$$

Equation of QF is $y = x - 2$ & PF is $x + y = 2$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (D)

$$\begin{aligned}S_n &= 1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^{n-1}} \\&= 2 \left(1 - \frac{1}{2^n} \right) = 2 - \frac{1}{2^{n-1}} \\ \text{Centre of } C_n &\text{ is } \left(2 - \frac{1}{2^{n-2}}, 0 \right)\end{aligned}$$

and radius of C_n is $\frac{1}{2^{n-1}}$

$$\text{when } r = \frac{1025}{513} < 2$$

C_n will lie inside m

$$\text{when } 2 - \frac{1}{2^{n-2}} + \frac{1}{2^{n-1}} < \frac{1025}{513}$$

$$\Rightarrow k = 10$$

$$\text{Also } \ell = 5$$

$$3k + 2\ell = 30 + 10 = 40$$

8. Ans (B)

Center of D_n is (S_{n-1}, S_{n-1})

$$r = \frac{1}{2^{n-1}}$$

D_n will lie inside

$$\text{when } \sqrt{2} (S_{n-1}) < \frac{2^{199} - 1}{2^{198}} \sqrt{2}$$

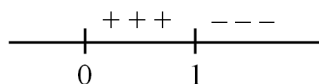
$$\Rightarrow \frac{\sqrt{2}}{2^{n-2}} > \frac{\sqrt{2}}{2^{198}} + \frac{1}{2^{n-1}}$$

$$\Rightarrow n = 199$$

9. Ans (C)

$$f'(x) = (|x| - x^2)e^{-x^2} + (|x| - x^2)e^{-x^2}, \quad x \geq 0$$

$$f' = 2(x - x^2)e^{-x^2}$$



hence option (D) is wrong

$$g'(x) = xe^{-x^2} 2x$$

$$f'(x) + g'(x) = 2 \times e^{-x^2}$$

$$f(x) + g(x) = -e^{-x^2} + c$$

$$f(x) + g(x) = -e^{-x^2} + 1$$

$$F(\ln 3) + g(\sqrt{\ln 3}) = 1 - \frac{1}{3} = \frac{2}{3} \quad (\text{option (A)})$$

is wrong)

$$H(x) = \psi_1(x) - 1 - \alpha x = e^{-x} + x - 1 - \alpha x, \quad x \geq 1 \text{ \& } \alpha \in (1, x)$$

$$H(1) = e^{-1} + 1 - 1 - \alpha < 0$$

$H'(x) = -e^{-x} + 1 - \alpha > 0 \Rightarrow H(x)$ is $\downarrow \Rightarrow$ option (B) is wrong

$$(C) \psi_2(x) = 2(\psi_1(\beta) - 1)$$

Applying L.M.V.T to $\psi_2(x)$ in $[0, x]$

$$\psi_2'(\beta) = \frac{\psi_2(x) - \psi_2(0)}{x}$$

$$2\beta - 2 + 2e^{-\beta} = \frac{\psi_2(x) - 0}{x}$$

$\Rightarrow \psi_2(x) = 2x(\psi_1(\beta) - 1)$ has one solution

10. Ans (D)

$$(A) \psi_1(x) = e^{-x} + x, \quad x \geq 0$$

$$\psi_1'(x) = 1 - e^{-x} > 0 \Rightarrow \psi_1(x) \text{ is } \uparrow$$

$$\psi_1(x) \geq \psi_1(0) \quad \forall x \geq 0 \Rightarrow \psi_1(x) \geq 1$$

$$(B) \psi_2(x) = x^2 - 2x + 2 - 2e^{-x} \quad x \geq 0$$

$$\psi_2'(x) = 2x - 2 + 2e^{-x} = 2\psi_1(x) - 2 \geq 0 \quad \forall x \geq 0$$

$\Rightarrow \psi_2(x)$ is

$$\uparrow \Rightarrow \psi_2(x) \geq \psi_2(0) \Rightarrow \psi_2(x) \geq 0$$

$$(C) f(x) = 2 \int_0^x (t - t^2)e^{-t^2} dt \text{ \& } x \in \left(0, \frac{1}{2}\right)$$

$$= \int_0^x 2te^{-t^2} dt - \int_0^x 2t^2e^{-t^2} dt$$

$$= -e^{-x^2} \Big|_0^x$$

$$\text{Let } H(x) = f(x) - 1 +$$

$$e^{-x^2} + \frac{2}{3}x^3 - \frac{2}{5}x^5, \quad x \in \left(0, \frac{1}{2}\right)$$

$$H(0) = 0$$

$$H'(x) = 2(x - x^2)e^{-x^2} - 2xe^{-x^2} + 2x^2 - 2x^4$$

$$= -2x^2e^{-x^2} + 2x^2 - 2x^4$$

$$= 2x^2(1 - x^2 - e^{-x^2})$$

$$\because e^{-x} \geq 1 - x \quad \forall x \geq 0$$

$$\Rightarrow H'(x) \leq 0$$

$$\Rightarrow H(x) \text{ is } \downarrow \Rightarrow 1 - 1(x) < 0 \quad \forall x \in \left(0, \frac{1}{2}\right)$$

$$\text{Let } P(x) = g(x)$$

$$-\frac{2}{3}x^3 + \frac{2}{5}x^5 - \frac{1}{7}x^7 \in \left(0, \frac{1}{2}\right)$$

$$P'(x) = 2x^2e^{-x^2} - 2x^2 + 2x^4 - x^6$$

$$= 2x^2 \left(1 - \frac{x^2}{1} + \frac{x^4}{2} - \frac{x^6}{3} + \dots\right) - 2x^2 + 2x^4 - x^6$$

$$= -\frac{x^8}{3} + \frac{x^{10}}{12} \dots$$

$$\Rightarrow P'(x) \leq 0$$

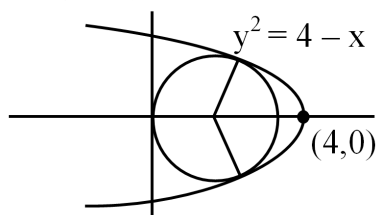
$$\Rightarrow P(x) \text{ is } \downarrow$$

$$\Rightarrow P(x) \leq 0$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (1.50)



Let the circle be

$$x^2 + y^2 + \lambda x = 0$$

For point of intersection of circle & parabola $y^2 = 4 - x$.

$$x^2 + 4 - x + \lambda x = 0 \Rightarrow x^2 + x(\lambda - 1) + 4 = 0$$

For tangency : $\Delta = 0 \Rightarrow (\lambda - 1)^2 - 16 = 0 \Rightarrow \lambda = 5$ (rejected) or $\lambda = -3$

$$\text{Circle : } x^2 + y^2 - 3x = 0$$

$$\text{Radius} = \frac{3}{2} = 1.5$$

2. Ans (2.00)

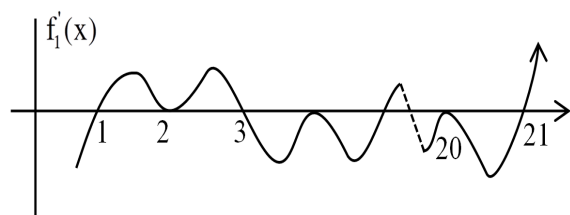
For point of intersection :

$$x^2 - 4x + 4 = 0 \Rightarrow x = 2 \text{ so } \alpha = 2$$

3. Ans (57.00)

$$f_1(x) = \int_0^x \prod_{j=1}^{21} (t - j)^j dt$$

$$f_1'(x) = \prod_{j=1}^{21} (x - j)^j = (x-1)(x-2)^2(x-3)^3 \dots (x-21)^{21}$$



So points of minima are $4m + 1$ where $m = 0, 1, \dots, 5 \Rightarrow m_1 = 6$

Points of maxima are $4m - 1$ where $m = 1, 2, \dots, 5 \Rightarrow n_1 = 5$
 $\Rightarrow 2m_1 + 3n_1 + m_1 n_1 = 57$

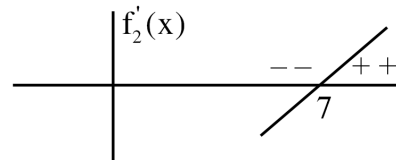
4. Ans (6.00)

$$f_2'(x) = 98(x-1)^{50} - 600(x-1)^{49} + 2450$$

$$\Rightarrow f_2'(x) = 2 \times 49 \times 50(x-1)^{49} - 50 \times 12 \times 49(x-1)^{48}$$

$$= 50 \times 49 \times 2(x-1)^{48}(x-1-6)$$

$$= 50 \times 49 \times 2(x-1)^{48}(x-7)$$



Point of minima = 7

$$\Rightarrow m_2 = 1$$

No point of maxima

$$\Rightarrow n_2 = 0$$

$$6m_2 + 4n_2 + 8m_2 n_2 = 6$$

5. Ans (2.00)

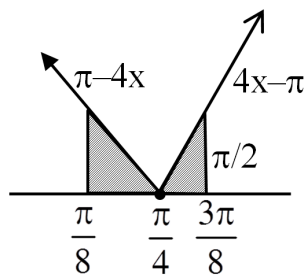
$$S_1 = \int_{\pi/8}^{3\pi/8} f(x) dx = \int_{\pi/8}^{3\pi/8} \sin^2 x dx = \int_{\pi/8}^{3\pi/8} \sin^2 \left(\frac{\pi}{8} + \frac{3\pi}{8} - x \right) dx$$

$$= \int_{\pi/8}^{3\pi/8} \cos^2 x dx$$

$$2S_1 = \int_{\pi/8}^{3\pi/8} (\sin^2 x + \cos^2 x) dx = \frac{3\pi}{8} - \frac{\pi}{8} = \frac{\pi}{4}$$

$$\Rightarrow \frac{16S_1}{\pi} = 2$$

6. Ans (1.50)



$$\begin{aligned}
 S_2 &= \int_{\pi/8}^{3\pi/8} f(x)g_2(x)dx = \int_{\pi/8}^{3\pi/8} \sin^2 x |4x - \pi| dx \\
 &= \int_{\pi/8}^{3\pi/8} \sin^2 \left(\frac{\pi}{2} - x \right) \left| 4 \left(\frac{\pi}{2} - x \right) - \pi \right| dx \\
 &= \int_{\pi/8}^{3\pi/8} (\cos^2 x) |\pi - 4x| dx \\
 \Rightarrow 2S_2 &= \int_{\pi/8}^{3\pi/8} |4x - \pi| (\sin^2 x + \cos^2 x) dx \\
 &= \int_{\pi/8}^{3\pi/8} |4x - \pi| dx \\
 &= 2 \times \frac{1}{2} \times \frac{\pi}{8} \times \frac{\pi}{2} = \frac{\pi^2}{16} \\
 \Rightarrow \frac{48S_2}{\pi^2} &= \frac{3}{2} = 1.5
 \end{aligned}$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (214)

A = set of numbers divisible by 3

$$A = \{3, 6, 9, 12, \dots, 1998\}$$

$$\therefore n(A) = 666$$

B = set of numbers divisible by 7

$$B = \{7, 14, 21, \dots, 1995\}$$

$$\therefore n(B) = 285$$

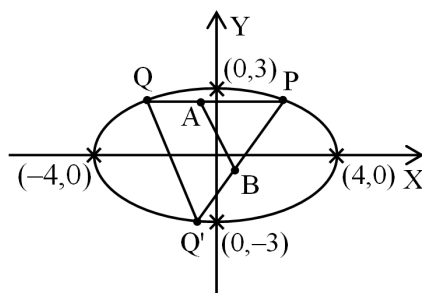
$$A \cap B = \{21, 42, \dots, 1995\}$$

$$\therefore n(A \cup B) = 606 + 285 - 95 = 856$$

$$\text{required probability} = \frac{856}{2000} = P$$

$$\text{so, } 500P = \frac{856}{2000} \times 500 = 214$$

8. Ans (4)



A and B be midpoints of segment PQ and PQ' respectively

$$AB = \text{distance between } M(P, Q) \text{ and } M(P, Q') = \frac{1}{2} \cdot QQ'$$

Since, Q, Q' must be on E, so,
maximum of QQ' = 8

$$\therefore \text{Maximum of } AB = \frac{8}{2} = 4$$

9. Ans (182)

$$\text{Let } f(x) = \left(\frac{10x}{x+1} \right)$$

$$\text{So, } f'(x) = 10 \left(\frac{(x+1) - x}{(x+1)^2} \right) = \frac{10}{(x+1)^2} > 0$$

$$\forall x \in [0, 10],$$

So, f(x) is an increasing function

$$\text{So, range of } f(x) \text{ is } \left[0, \sqrt{\frac{100}{11}} \right]$$

$$\begin{aligned}
 I &= \int_0^{1/9} \left[\sqrt{\frac{10x}{x+1}} \right] dx + \int_{1/9}^9 \left[\sqrt{\frac{10x}{x+1}} \right] dx + \int_9^{2/3} \left[\sqrt{\frac{10x}{x+1}} \right] dx + \int_{2/3}^9 \left[\sqrt{\frac{10x}{x+1}} \right] dx + \int_9^{10} \left[\sqrt{\frac{10x}{x+1}} \right] dx \\
 &= 0 + \int_{1/9}^{2/3} dx + 2 \int_{2/3}^9 dx + 3 \int_9^{10} dx \\
 &= \frac{2}{3} - \frac{1}{9} + 2 \left(9 - \frac{2}{3} \right) + 3(10 - 9) \\
 &= \frac{6-1}{9} + 2 \times \frac{25}{3} + 3 = \frac{5}{9} + \frac{50}{3} + 3 \\
 &= \frac{5+150+27}{9} = \frac{182}{9} = 182
 \end{aligned}$$